

Analysis II

Exercise 1.

Topologies. $\mathcal{E}(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$ with seminorms

$$p_{K,\beta}(f) = \sup_{x \in K} |\partial^\beta f(x)|, \quad K \Subset \mathbb{R}^n.$$

$\mathcal{S}(\mathbb{R}^n)$ with seminorms

$$q_{\alpha,\beta}(f) = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)|.$$

$\mathcal{D}(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n)$ with the LF topology: a sequence $f_j \rightarrow 0$ in $\mathcal{D}$ if and only if there exists a compact set $K \Subset \mathbb{R}^n$ such that $\text{supp } f_j \subset K$ for all sufficiently large j , and

$$\sup_{x \in K} |\partial^\beta f_j(x)| \rightarrow 0 \quad \text{for every multiindex } \beta.$$

(a) $\mathcal{S}(\mathbb{R}^n)$ is a vector subspace of $\mathcal{E}(\mathbb{R}^n)$ and $\mathcal{S} \rightarrow \mathcal{E}$

If $f \in \mathcal{S}$, then for each β , $q_{0,\beta}(f) = \sup_{\mathbb{R}^n} |\partial^\beta f| < \infty$, hence $f \in C^\infty(\mathbb{R}^n) = \mathcal{E}$. Linearity of $\mathcal{S}$ follows from linearity of the seminorms. If $f_j \rightarrow 0$ in $\mathcal{S}$, then for any compact K and any β ,

$$p_{K,\beta}(f_j) = \sup_{x \in K} |\partial^\beta f_j(x)| \leq \sup_{x \in \mathbb{R}^n} |\partial^\beta f_j(x)| = q_{0,\beta}(f_j) \xrightarrow{j \rightarrow \infty} 0,$$

so $f_j \rightarrow 0$ in $\mathcal{E}$.

(b) $\mathcal{D}(\mathbb{R}^n)$ is a vector subspace of $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{D} \rightarrow \mathcal{S}$

If $f \in C_c^\infty$ with $\text{supp } f \subset K$, then for any α, β ,

$$q_{\alpha,\beta}(f) = \sup_{x \in K} |x^\alpha \partial^\beta f(x)| \leq \left(\sup_{x \in K} |x^\alpha| \right) \cdot \sup_{x \in K} |\partial^\beta f(x)| < \infty,$$

hence $f \in \mathcal{S}$. Linearity is obvious. If $f_j \rightarrow 0$ in $\mathcal{D}$, choose a compact K containing $\text{supp } f_j$ for large j ; then

$$q_{\alpha,\beta}(f_j) = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f_j(x)| = \sup_{x \in K} |x^\alpha \partial^\beta f_j(x)| \leq \left(\sup_{x \in K} |x^\alpha| \right) \cdot \sup_{x \in K} |\partial^\beta f_j(x)| \xrightarrow{j \rightarrow \infty} 0,$$

so $f_j \rightarrow 0$ in $\mathcal{S}$.

(c) Sequences in $C_c^\infty(\mathbb{R}^n)$ with contrasting limits

(i) $\phi_j \rightarrow 0$ in $\mathcal{S}$ but not in $\mathcal{D}$. Fix $\psi \in C_c^\infty(\mathbb{R}^n)$, $\psi \not\equiv 0$, with $\text{supp } \psi \subset B(0, 1)$. Let $e_1 = (1, 0, \dots, 0)$ and define

$$\phi_j(x) := e^{-j^2} \psi(x - je_1).$$

For any α, β there is C_β with $\sup_x |x^\alpha \partial^\beta \phi_j(x)| \leq C_\beta e^{-j^2} (1 + j)^{|\alpha|} \rightarrow 0$; thus $\phi_j \rightarrow 0$ in $\mathcal{S}$. However $\text{supp } \phi_j \subset B(je_1, 1)$ escapes to infinity, so no compact K contains the supports eventually; therefore ϕ_j does not converge to 0 in $\mathcal{D}$.

(ii) $\phi_j \rightarrow 0$ in $\mathcal{E}$ but not in $\mathcal{S}$. With the same ψ , set

$$\phi_j(x) := \psi(x - je_1).$$

For any fixed compact K , $K \cap \text{supp } \phi_j = \emptyset$ for j large, hence for every β , $\sup_{x \in K} |\partial^\beta \phi_j(x)| = 0$. Thus $\phi_j \rightarrow 0$ in $\mathcal{E}$. On the other hand, for $\alpha \neq 0$ and $\beta = 0$,

$$q_{\alpha,0}(\phi_j) \geq \sup_{|y| \leq 1} |(je_1 + y)^\alpha| |\psi(y)| \gtrsim j^{|\alpha|} \rightarrow \infty,$$

so ϕ_j has no limit in $\mathcal{S}$.

Rapidly decreasing functions and the Schwartz space

Rapid decay. A smooth function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is called *rapidly decreasing* if for every pair of multiindices α, β one has

$$\sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)| < \infty \quad \text{and} \quad \lim_{|x| \rightarrow \infty} |x^\alpha \partial^\beta f(x)| = 0.$$

Equivalently, for each $N \in \mathbb{N}$ and each β there exists $C_{N,\beta}$ such that

$$|\partial^\beta f(x)| \leq C_{N,\beta} (1 + |x|)^{-N} \quad (\forall x \in \mathbb{R}^n).$$

Schwartz space. The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of all rapidly decreasing C^∞ functions and is a Fréchet space with seminorms

$$q_{\alpha,\beta}(f) := \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)|, \quad \alpha, \beta \in \mathbb{N}^n.$$

The Fourier transform is a topological automorphism of $\mathcal{S}(\mathbb{R}^n)$.

Fréchet topology (definition). A Fréchet space is a complete metrizable locally convex space; equivalently, its topology is induced by a countable family of seminorms $(p_k)_{k \in \mathbb{N}}$, and it is complete for the metric

$$d(x, y) = \sum_{k=1}^{\infty} 2^{-k} \frac{p_k(x - y)}{1 + p_k(x - y)}.$$

The spaces $\mathcal{E}(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ are Fréchet with the standard seminorm families

$$\mathcal{E} : p_{K,\beta}(f) = \sup_{x \in K} |\partial^\beta f(x)| \quad (K \Subset \mathbb{R}^n), \quad \mathcal{S} : q_{\alpha,\beta}(f) = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)|.$$

Examples and nonexamples. The Gaussian $x \mapsto e^{-|x|^2}$ and any compactly supported C^∞ function belong to $\mathcal{S}(\mathbb{R}^n)$. Functions like $(1 + |x|)^{-1}$ or $\sin |x|$ do not: the former decays too slowly, the latter does not decay.

Sobolev spaces (brief overview)

Integer order. For $k \in \mathbb{N}$ and $1 \leq p \leq \infty$ the Sobolev space is

$$W^{k,p}(\mathbb{R}^n) = \left\{ f \in L^p(\mathbb{R}^n) : \partial^\alpha f \in L^p \text{ for all } |\alpha| \leq k \right\}, \quad \|f\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^p}.$$

Fractional order. For $p = 2$ and $s \in \mathbb{R}$ one defines the Bessel potential spaces

$$H^s(\mathbb{R}^n) = \left\{ f \in \mathcal{S}'(\mathbb{R}^n) : (1 + |\xi|^2)^{s/2} \hat{f}(\xi) \in L^2(\mathbb{R}^n) \right\}, \quad \|f\|_{H^s}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi.$$

Relations. One has $\mathcal{S}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$ for all $s \in \mathbb{R}$, and $\mathcal{S}(\mathbb{R}^n)$ is dense in $W^{k,p}(\mathbb{R}^n)$ for $1 \leq p < \infty$. The dual $\mathcal{S}'(\mathbb{R}^n)$ is the space of tempered distributions.

The LF topology on $\mathcal{D}(\mathbb{R}^n)$

Inductive limit description. For a compact set $K \Subset \mathbb{R}^n$ define

$$\mathcal{D}(K) = \{ f \in C^\infty(\mathbb{R}^n) : \text{supp } f \subset K \}, \quad \|f\|_{K,m} := \max_{|\beta| \leq m} \sup_{x \in K} |\partial^\beta f(x)| \quad (m \in \mathbb{N}).$$

Each $\mathcal{D}(K)$ is a Fréchet space with the seminorms $\|\cdot\|_{K,m}$. Fix an exhaustion by compacts $K_1 \Subset K_2 \Subset \cdots$ with $\bigcup_{m \geq 1} K_m = \mathbb{R}^n$; then

$$\mathcal{D}(\mathbb{R}^n) = \bigcup_{m \geq 1} \mathcal{D}(K_m)$$

endowed with the finest locally convex topology making the inclusions $\mathcal{D}(K_m) \hookrightarrow \mathcal{D}(\mathbb{R}^n)$ continuous. This locally convex space is the strict inductive limit (LF space) of the Fréchet spaces $\mathcal{D}(K_m)$.

Convergence criterion (sequences). A sequence f_j converges to 0 in $\mathcal{D}(\mathbb{R}^n)$ if and only if there exists a compact $K \Subset \mathbb{R}^n$ such that $\text{supp } f_j \subset K$ for all sufficiently large j , and for every multiindex β one has

$$\sup_{x \in K} |\partial^\beta f_j(x)| \xrightarrow{j \rightarrow \infty} 0.$$

Equivalently, all derivatives of f_j tend to 0 uniformly on a fixed compact set that eventually contains the supports.

Basic properties and density. The space $\mathcal{D}(\mathbb{R}^n)$ is complete, nuclear and Montel; it is neither normable nor metrizable when $\mathbb{R}^n$ is non-compact. One has the continuous dense embeddings $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset \mathcal{E}(\mathbb{R}^n)$, and $\mathcal{D}(\mathbb{R}^n)$ is dense in both $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{E}(\mathbb{R}^n)$. For density in $\mathcal{E}$ on $\mathbb{R}^n$, given $f \in C^\infty$ and a compact K , choose $\chi_R \in C_c^\infty$ with $\chi_R \equiv 1$ on K and $\text{supp } \chi_R \subset B(0, R)$; then $\chi_R f \in \mathcal{D}$ and for each multiindex β ,

$$\sup_{x \in K} |\partial^\beta (\chi_R f - f)(x)| = 0 \quad \text{for } R \text{ sufficiently large,}$$

so $\chi_R f \rightarrow f$ in $\mathcal{E}$. For density in $\mathcal{S}$ one first cut off f at infinity and then mollify to achieve rapid decay of all derivatives.

Problem 2 — Stability of translations and difference quotients

Let

$$X \in \{\mathcal{D}(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n), \mathcal{E}(\mathbb{R}^n)\}, \quad \phi \in X.$$

Establish the following properties.

(a) If $x_\ell \in \mathbb{R}^n$ with $x_\ell \rightarrow 0$, then

$$\tau_{x_\ell} \phi \longrightarrow \phi \quad \text{in } X \text{ as } \ell \rightarrow \infty,$$

where the translation operator τ_x is defined by

$$(\tau_x \phi)(y) := \phi(y - x).$$

(b) If $h_\ell \in \mathbb{R}$ with $h_\ell \rightarrow 0$, then

$$\Delta_i^{h_\ell} \phi \longrightarrow D_i \phi \quad \text{in } X \text{ as } \ell \rightarrow \infty,$$

where

$$\Delta_i^h \phi := h^{-1} (\tau_{he_i} \phi - \phi) = \frac{\phi(\cdot - he_i) - \phi}{h}$$

is the *difference quotient* in the i -th coordinate direction.

In words: small translations of a smooth function tend to zero in the topology of X , and the difference quotients converge to the partial derivative $D_i\phi$ in the same sense.

Proof of (a). *Case $X = \mathcal{E}(\mathbb{R}^n)$.* Fix a compact $K \Subset \mathbb{R}^n$ and $m \in \mathbb{N}$. For any multi-index α with $|\alpha| \leq m$,

$$\sup_{y \in K} |D^\alpha(\tau_{x_\ell}\phi - \phi)(y)| = \sup_{y \in K} |D^\alpha\phi(y - x_\ell) - D^\alpha\phi(y)|.$$

Choose a compact K' containing $K \cup (K - x_\ell)$ for all large ℓ . Since $D^\alpha\phi$ is uniformly continuous on K' , the right-hand side tends to 0 as $\ell \rightarrow \infty$. Hence $\tau_{x_\ell}\phi \rightarrow \phi$ in $\mathcal{E}$.

Case $X = \mathcal{D}(\mathbb{R}^n)$. Choose a compact K with $\text{supp } \phi \subset K^\circ$. For ℓ large,

$$\text{supp}(\tau_{x_\ell}\phi) \subset K + x_\ell \subset K'',$$

for some fixed compact K'' . Applying the estimate from the $\mathcal{E}$ case on K'' yields, for every m ,

$$\sup_{y \in K''} |D^\alpha(\tau_{x_\ell}\phi - \phi)(y)| \xrightarrow{\ell \rightarrow \infty} 0 \quad \text{for all } |\alpha| \leq m,$$

and the supports remain in the common compact K'' . Therefore $\tau_{x_\ell}\phi \rightarrow \phi$ in $\mathcal{D}$.

Case $X = \mathcal{S}(\mathbb{R}^n)$. For any α, β ,

$$x^\alpha D^\beta(\tau_{x_\ell}\phi - \phi)(x) = (x^\alpha - (x - x_\ell)^\alpha) D^\beta\phi(x - x_\ell) + (x - x_\ell)^\alpha (D^\beta\phi(x - x_\ell) - D^\beta\phi(x)).$$

The polynomial difference satisfies

$$|x^\alpha - (x - x_\ell)^\alpha| \leq C_\alpha |x_\ell| (1 + |x|)^{|\alpha|-1},$$

and all derivatives of ϕ are rapidly decreasing; hence the first term tends to 0 uniformly in x . For the second term, the uniform continuity of $D^\beta\phi$ together with rapid decrease implies that, for every N ,

$$\sup_{x \in \mathbb{R}^n} (1 + |x|)^{-N} |x^\alpha D^\beta(\tau_{x_\ell}\phi - \phi)(x)| \longrightarrow 0 \quad (\ell \rightarrow \infty).$$

Thus every Schwartz seminorm goes to 0, and $\tau_{x_\ell}\phi \rightarrow \phi$ in $\mathcal{S}(\mathbb{R}^n)$.

Proof of (b). Fix i and a multi-index α . For any $h \neq 0$,

$$D^\alpha \Delta_i^h \phi = \frac{D^\alpha \phi(\cdot - he_i) - D^\alpha \phi}{h}.$$

Case $X = \mathcal{E}$ (and inside a fixed $\mathcal{D}_K \subset \mathcal{D}$). Let $K \Subset \mathbb{R}^n$. By the mean value theorem along e_i ,

$$\frac{D^\alpha \phi(x - he_i) - D^\alpha \phi(x)}{h} - D^{\alpha+e_i} \phi(x) = \int_0^1 \left(D^{\alpha+e_i} \phi(x - the_i) - D^{\alpha+e_i} \phi(x) \right) dt.$$

Hence

$$\sup_{x \in K} \left| D^\alpha \Delta_i^h \phi(x) - D^{\alpha+e_i} \phi(x) \right| \leq \sup_{0 \leq t \leq 1} \sup_{x \in K} \left| D^{\alpha+e_i} \phi(x - the_i) - D^{\alpha+e_i} \phi(x) \right| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

For $\mathcal{D}$, the supports remain in a common compact set for $|h|$ small, so the same uniform estimate yields convergence in $\mathcal{D}$.

Case $X = \mathcal{S}$. For any α, β ,

$$x^\beta \left(D^\alpha \Delta_i^h \phi - D^{\alpha+e_i} \phi \right)(x) = A_{\alpha, \beta, h}(x) + B_{\alpha, \beta, h}(x),$$

where

$$A_{\alpha, \beta, h}(x) = \frac{x^\beta - (x - he_i)^\beta}{h}$$

$$B_{\alpha, \beta, h}(x) = (x - he_i)^\beta \left(\frac{D^\alpha \phi(x - he_i) - D^\alpha \phi(x)}{h} - D^{\alpha+e_i} \phi(x) \right).$$

For $A_{\alpha, \beta, h}$, expand the polynomial difference:

$$\frac{x^\beta - (x - he_i)^\beta}{h} = \sum_{|\gamma| \leq |\beta| - 1} c_{\beta, \gamma}(h) x^\gamma, \quad |c_{\beta, \gamma}(h)| \leq C_\beta,$$

so that

$$|A_{\alpha, \beta, h}(x)| \leq C_\beta (1 + |x|)^{|\beta| - 1} |D^\alpha \phi(x - he_i)|.$$

Since $D^\alpha \phi$ is rapidly decreasing, the right-hand side defines a family bounded in every Schwartz seminorm and tends to 0 as $h \rightarrow 0$.

For $B_{\alpha, \beta, h}$, the integral remainder form of the mean value theorem gives

$$\frac{D^\alpha \phi(x - he_i) - D^\alpha \phi(x)}{h} - D^{\alpha+e_i} \phi(x) = \int_0^1 \left(D^{\alpha+e_i} \phi(x - the_i) - D^{\alpha+e_i} \phi(x) \right) dt,$$

whence

$$|B_{\alpha, \beta, h}(x)| \leq (1 + |x|)^{|\beta|} \int_0^1 \left| D^{\alpha+e_i} \phi(x - the_i) - D^{\alpha+e_i} \phi(x) \right| dt.$$

The integrand tends to 0 uniformly in x by the uniform continuity of $D^{\alpha+e_i}\phi$, and the polynomial factor is absorbed by the rapid decay of the derivatives of ϕ . Consequently, for every pair (α, β) ,

$$q_{\alpha,\beta}(\Delta_i^h \phi - D_i \phi) \longrightarrow 0 \quad (h \rightarrow 0).$$

This proves $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{E}$, $\mathcal{D}$, and $\mathcal{S}$.

Example 1. Let $\phi(x) = e^{-|x|^2} \in \mathcal{S}(\mathbb{R}^n)$. Then, for any translation vector $x_\ell \rightarrow 0$,

$$(\tau_{x_\ell} \phi)(y) = e^{-|y-x_\ell|^2} \longrightarrow e^{-|y|^2} = \phi(y)$$

together with all derivatives, because every derivative of ϕ is a polynomial times $e^{-|y|^2}$, which is uniformly continuous and rapidly decreasing. Hence each Schwartz seminorm

$$q_{\alpha,\beta}(\tau_{x_\ell} \phi - \phi) = \sup_{y \in \mathbb{R}^n} |y^\alpha D^\beta (\tau_{x_\ell} \phi - \phi)(y)|$$

tends to 0 as $\ell \rightarrow \infty$.

For the difference quotient,

$$\Delta_i^h \phi(x) = \frac{\phi(x - he_i) - \phi(x)}{h},$$

Taylor's theorem gives

$$\phi(x - he_i) = \phi(x) - h D_i \phi(x) + \frac{h^2}{2} D_i^2 \phi(x - \theta he_i), \quad 0 < \theta < 1,$$

so that

$$\Delta_i^h \phi(x) - D_i \phi(x) = \frac{h}{2} D_i^2 \phi(x - \theta he_i) \rightarrow 0$$

uniformly with all derivatives. Therefore $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{S}(\mathbb{R}^n)$.

Example 2. Let $\phi \in \mathcal{D}(\mathbb{R})$ be compactly supported in $[-1, 1]$. For small $|h|$,

$$\text{supp}(\phi(\cdot - h)) \subset [-1 + h, 1 + h] \subset [-2, 2].$$

Hence every difference quotient

$$\Delta^h \phi(x) = \frac{\phi(x - h) - \phi(x)}{h}$$

has support contained in a fixed compact interval, and all derivatives converge uniformly:

$$\sup_{x \in [-2, 2]} |D^m (\Delta^h \phi - \phi')(x)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Thus $\Delta^h \phi \rightarrow \phi'$ in the $\mathcal{D}$ -topology.

Problem 3 — Distributions with zero derivative are constants

Statement. Suppose $u \in \mathcal{D}'(\mathbb{R})$ satisfies $Du = 0$. Show that there exists $\lambda \in \mathbb{C}$ such that

$$u[\phi] = \lambda \int_{\mathbb{R}} \phi(x) dx, \quad \forall \phi \in \mathcal{D}(\mathbb{R}).$$

(Extend the result to $\mathbb{R}^n$ for $n > 1$).

Proof. The hypothesis $Du = 0$ means $u[\phi'] = 0$ for all $\phi \in \mathcal{D}(\mathbb{R})$. Fix $\phi_0 \in \mathcal{D}(\mathbb{R})$ with $\int_{\mathbb{R}} \phi_0 = 1$. For any $\phi \in \mathcal{D}(\mathbb{R})$, set $c_\phi = \int_{\mathbb{R}} \phi$ and define

$$\psi(x) = \int_{-\infty}^x (\phi(t) - c_\phi \phi_0(t)) dt.$$

Then $\psi \in \mathcal{D}(\mathbb{R})$ and $\psi' = \phi - c_\phi \phi_0$. Therefore

$$u[\phi] = u[\psi'] + c_\phi u[\phi_0] = c_\phi u[\phi_0] = \lambda \int_{\mathbb{R}} \phi,$$

with $\lambda := u[\phi_0]$.

Extension to $\mathbb{R}^n$. If $D_j u = 0$ for all $j = 1, \dots, n$, then $u[\partial_j \phi] = 0$ for all $\phi \in \mathcal{D}(\mathbb{R}^n)$. Choose $\phi_0 \in \mathcal{D}(\mathbb{R}^n)$ with $\int \phi_0 = 1$.

For any ϕ , write $\phi = \nabla \cdot F + c_\phi \phi_0$ with $c_\phi = \int \phi$ and $F \in \mathcal{D}(\mathbb{R}^n; \mathbb{C}^n)$ such that $\nabla \cdot F = \phi - c_\phi \phi_0$ (possible because the right-hand side has integral 0).

Then

$$u[\phi] = u[\nabla \cdot F] + c_\phi u[\phi_0] = \lambda \int_{\mathbb{R}^n} \phi.$$

Examples. If u is induced by $f \in L^1_{\text{loc}}(\mathbb{R})$, then $Du = 0$ implies $f' = 0$ in the distributional sense, hence f is a.e. constant and $u[\phi] = \lambda \int \phi$. By contrast, $D\delta \neq 0$, since $D\delta[\phi] = -\phi'(0)$.

Definition (weak-* topology on X').

Let X be a locally convex space over $\mathbb{K}$ and let X' denote its continuous dual. The weak-* topology on X' , denoted $\sigma(X', X)$, is the coarsest topology on X' for which every evaluation map

$$e_x : X' \rightarrow \mathbb{K}, \quad e_x(f) = f(x) \quad (x \in X),$$

is continuous. Equivalently, it is the initial topology induced by the family $\{e_x : x \in X\}$.

Convergence characterization. A net $(f_\alpha) \subset X'$ converges to $f \in X'$ in $\sigma(X', X)$ if and only if

$$f_\alpha(x) \longrightarrow f(x) \quad \text{for every } x \in X.$$

Neighborhoods and subbasis. For $x \in X$ and an open set $U \subset \mathbb{K}$, the sets

$$e_x^{-1}(U) = \{ f \in X' : f(x) \in U \}$$

form a subbasis. A typical 0-neighborhood is

$$V(x_1, \dots, x_m; \varepsilon) = \left\{ f \in X' : |f(x_j)| < \varepsilon \text{ for } j = 1, \dots, m \right\}.$$

Basic properties. If X is normed, the closed unit ball of X' is compact in $\sigma(X', X)$ (Banach–Alaoglu). If X is separable, the weak-* topology is metrizable on norm-bounded subsets of X' . The weak topology on X is $\sigma(X, X')$, while the weak-* topology on X' is $\sigma(X', X)$. The canonical embedding $J : X \rightarrow X''$ is weak-to-weak-* continuous; it is onto if and only if X is reflexive.

Examples. For $X = \ell^1$ and $X' = \ell^\infty$, $\sigma(\ell^\infty, \ell^1)$ is pointwise convergence on ℓ^1 : $y^{(\alpha)} \rightarrow y$ iff $\sum_k y_k^{(\alpha)} x_k \rightarrow \sum_k y_k x_k$ for all $x \in \ell^1$.

For $X = C_0(\Omega)$ (locally compact Hausdorff), $X' = M(\Omega)$; the topology $\sigma(M(\Omega), C_0(\Omega))$ is narrow convergence of measures: $\int f d\mu_\alpha \rightarrow \int f d\mu$ for all $f \in C_0(\Omega)$.

For distributions, $\sigma(\mathcal{D}'(\mathbb{R}^n), \mathcal{D}(\mathbb{R}^n))$ and $\sigma(\mathcal{S}'(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n))$ coincide with test-functionwise convergence.

Problem 4 — Weak-* limit of difference quotients of distributions

Statement. Let $X \in \{\mathcal{D}(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n), \mathcal{E}(\mathbb{R}^n)\}$. For $u \in X'$ and $x \in \mathbb{R}^n$ define $\tau_x u$ by $(\tau_x u)[\phi] = u[\tau_{-x}\phi]$ for all $\phi \in X$. Let $\Delta_i^h u := h^{-1}(\tau_{he_i} u - u)$. Show that $\Delta_i^h u \rightarrow D_i u$ as $h \rightarrow 0$ in the weak-* topology $\sigma(X', X)$.

Proof. For any $\phi \in X$,

$$(\Delta_i^h u)[\phi] = \frac{u[\tau_{-he_i}\phi] - u[\phi]}{h} = u\left[\frac{\tau_{-he_i}\phi - \phi}{h}\right] = -u\left[\frac{\tau_{he_i}\phi - \phi}{h}\right] = -u[\Delta_i^h \phi].$$

By Problem 2(b), $\Delta_i^h \phi \rightarrow D_i \phi$ in X . Since $u \in X'$ is continuous,

$$(\Delta_i^h u)[\phi] = -u[\Delta_i^h \phi] \longrightarrow -u[D_i \phi] = D_i u[\phi],$$

where $D_i u[\phi] := -u[D_i \phi]$. Hence $\Delta_i^h u \rightarrow D_i u$ in $\sigma(X', X)$.

Examples. Example 1 (regular distributions). Let $u = u_f$ be induced by $f \in L_{\text{loc}}^1(\mathbb{R}^n)$, i.e. $u_f[\phi] = \int_{\mathbb{R}^n} f(x) \phi(x) dx$. Then

$$(\Delta_i^h u_f)[\phi] = -u_f[\Delta_i^h \phi] = - \int_{\mathbb{R}^n} f(x) \frac{\phi(x - he_i) - \phi(x)}{h} dx.$$

If $f \in W_{\text{loc}}^{1,1}$ one may integrate by parts:

$$(\Delta_i^h u_f)[\phi] = \int_{\mathbb{R}^n} \frac{f(x) - f(x + he_i)}{h} \phi(x) dx \longrightarrow - \int_{\mathbb{R}^n} (\partial_i f)(x) \phi(x) dx = D_i u_f[\phi].$$

In general (for $f \in L_{\text{loc}}^1$ only), the limit still equals $D_i u_f[\phi]$ because $\Delta_i^h \phi \rightarrow D_i \phi$ in X and u_f is continuous on X (here $X = \mathcal{D}, \mathcal{S}, \mathcal{E}$ accordingly).

Example 2 (Dirac mass). For the Dirac distribution,

$$(\Delta_i^h \delta)[\phi] = \frac{\delta[\tau_{-he_i} \phi] - \delta[\phi]}{h} = \frac{\phi(he_i) - \phi(0)}{h} \longrightarrow \partial_i \phi(0) = -D_i \delta[\phi].$$

Hence $\Delta_i^h \delta \rightarrow D_i \delta$ in $\sigma(X', X)$, as required.

Example 3 (finite sums of shifted deltas). Let $u = \sum_{j=1}^m c_j \delta_{a_j}$ with fixed points $a_j \in \mathbb{R}^n$. Then

$$(\Delta_i^h u)[\phi] = \sum_{j=1}^m c_j \frac{\phi(a_j + he_i) - \phi(a_j)}{h} \longrightarrow \sum_{j=1}^m c_j \partial_i \phi(a_j) = - \sum_{j=1}^m c_j D_i \delta_{a_j}[\phi],$$

so $\Delta_i^h u \rightarrow \sum_j c_j D_i \delta_{a_j}$ weak- * .

Examples. If u is induced by $f \in L_{\text{loc}}^1$, then $(\Delta_i^h u)[\phi] = - \int \Delta_i^h \phi f \rightarrow - \int D_i \phi f = D_i u[\phi]$. For $u = \delta$, $\Delta_i^h \delta = (\delta(\cdot - he_i) - \delta)/h \rightarrow D_i \delta$ weak- * .

Problem 5 — Distributions annihilated by a power of

x

Statement. Suppose $u \in \mathcal{D}'(\mathbb{R})$ satisfies $xu = 0$. Show that $u = c\delta_0$ for some $c \in \mathbb{C}$. Find the most general $u \in \mathcal{D}'(\mathbb{R})$ such that $x^k u = 0$ for some $k \in \mathbb{N}$.

Proof. Since $\langle xu, \phi \rangle = \langle u, x\phi \rangle$, the condition $xu = 0$ says u vanishes on $x\mathcal{D}(\mathbb{R})$. Thus $\text{supp } u \subset \{0\}$ and $u = \sum_{m=0}^M c_m \delta_0^{(m)}$. Moreover,

$$x\delta_0^{(m)} = -m\delta_0^{(m-1)} \quad (m \geq 1), \quad x\delta_0 = 0.$$

Hence $0 = xu = \sum_{m=1}^M c_m(-m)\delta_0^{(m-1)}$ forces $c_m = 0$ for $m \geq 1$, so $u = c_0\delta_0$.

Iterating,

$$x^k\delta_0^{(m)} = (-1)^k m(m-1)\cdots(m-k+1)\delta_0^{(m-k)},$$

which is zero iff $m < k$. Therefore

$$\{u \in \mathcal{D}'(\mathbb{R}) : x^k u = 0\} = \text{span}\{\delta_0, \delta'_0, \dots, \delta_0^{(k-1)}\}.$$

Examples. Example 1 ($xu = 0$ forces a pure mass at 0). For $u = c_0\delta_0$,

$$\langle xu, \phi \rangle = \langle u, x\phi \rangle = c_0 x\phi(0) = 0,$$

so $xu = 0$. Conversely, if $xu = 0$ and $u = \sum_{m=0}^M c_m \delta_0^{(m)}$, then using $x\delta_0^{(m)} = -m\delta_0^{(m-1)}$,

$$0 = xu = \sum_{m=1}^M c_m(-m)\delta_0^{(m-1)} \Rightarrow c_m = 0 \quad (m \geq 1),$$

hence $u = c_0\delta_0$.

Example 2 (all solutions of $x^2u = 0$). Write $u = a\delta_0 + b\delta'_0 + \sum_{m \geq 2} c_m \delta_0^{(m)}$. Since $x^2\delta_0 = 0$ and $x^2\delta'_0 = x(-\delta_0) = 0$, while

$$x^2\delta_0^{(m)} = x(-m\delta_0^{(m-1)}) = m(m-1)\delta_0^{(m-2)} \neq 0 \quad \text{for } m \geq 2,$$

we must have $c_m = 0$ for $m \geq 2$. Therefore the solution space is $\text{span}\{\delta_0, \delta'_0\}$.

Example 3 (general k). Using $x^k\delta_0^{(m)} = (-1)^k m(m-1)\cdots(m-k+1)\delta_0^{(m-k)}$, the product vanishes iff $m < k$. Hence precisely the linear span

$$\{u : x^k u = 0\} = \text{span}\{\delta_0, \delta'_0, \dots, \delta_0^{(k-1)}\}.$$

Example 4 (failure when $m \geq k$). Take $k = 3$ and $u = \delta_0^{(3)}$. Then $x^3u = (-1)^3 3 \cdot 2 \cdot 1 \delta_0 = -6\delta_0 \neq 0$, so u is not a solution.

Examples. If $x^2u = 0$ then $u = a\delta_0 + b\delta'_0$. If $xu = 0$ then $u = c\delta_0$.

Problem 6 — Continuity criterion on $\mathcal{S}(\mathbb{R}^n)$

Statement. Let $u : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$ be linear. Show that u is continuous if and only if there exist $N, k \in \mathbb{N}$ and $C > 0$ such that

$$|u[\phi]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} (1 + |x|)^N |D^\alpha \phi(x)| \quad \text{for all } \phi \in \mathcal{S}(\mathbb{R}^n).$$

Proof. The topology of $\mathcal{S}(\mathbb{R}^n)$ is generated by the seminorms

$$p_{N,k}(\phi) = \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} (1 + |x|)^N |D^\alpha \phi(x)|.$$

($\Rightarrow$) If u is continuous at 0, there exist finitely many seminorms p_{N_j, k_j} and constants C_j with $|u[\phi]| \leq \sum_j C_j p_{N_j, k_j}(\phi)$. Let $N = \max_j N_j$, $k = \max_j k_j$, $C = \sum_j C_j$. Since $p_{N_j, k_j} \leq p_{N, k}$,

$$|u[\phi]| \leq C p_{N, k}(\phi).$$

($\Leftarrow$) Conversely, if $|u[\phi]| \leq C p_{N, k}(\phi)$ for all ϕ , then u is continuous because it is dominated by a continuous seminorm. This characterizes continuous linear functionals on $\mathcal{S}$ (tempered distributions).

Examples. Example 1 (integration against polynomially growing functions).

Let f satisfy $|f(x)| \leq C_f(1 + |x|)^M$. Define $u_f[\phi] = \int_{\mathbb{R}^n} f(x)\phi(x) dx$. Then for $N = M$ and $k = 0$,

$$|u_f[\phi]| \leq \int |f(x)| |\phi(x)| dx \leq C_f \int (1 + |x|)^N |\phi(x)| dx \leq C \sup_x (1 + |x|)^N |\phi(x)| = C p_{N, 0}(\phi),$$

where the last inequality uses that $\phi \in \mathcal{S}(\mathbb{R}^n)$ decays rapidly. Thus u_f is continuous.

Example 2 (derivatives of the Dirac delta). For $\beta \in \mathbb{N}^n$, $u = \partial^\beta \delta$ satisfies

$$|u[\phi]| = |\partial^\beta \phi(0)| \leq \sup_x \sum_{|\alpha| \leq |\beta|} |D^\alpha \phi(x)| \leq p_{0, |\beta|}(\phi).$$

Hence the criterion holds with $N = 0$, $k = |\beta|$.

Example 3 (moment functionals). Fix a multi-index γ . Define $m_\gamma[\phi] = \int_{\mathbb{R}^n} x^\gamma \phi(x) dx$. Since $|x^\gamma| \leq (1 + |x|)^{|\gamma|}$,

$$|m_\gamma[\phi]| \leq \int (1 + |x|)^{|\gamma|} |\phi(x)| dx \leq C p_{|\gamma|, 0}(\phi).$$

Thus m_γ is continuous with $N = |\gamma|$, $k = 0$.

Example 4 (finite order differential operators with polynomial coefficients).

Let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$, with $|a_\alpha(x)| \leq C(1+|x|)^M$. Define $u_P[\phi] = \int_{\mathbb{R}^n} (P(x, D)\phi)(x) dx$. Then by Leibniz' rule and the polynomial bound on a_α ,

$$|u_P[\phi]| \leq \sum_{|\alpha| \leq m} \int |a_\alpha(x)| |D^\alpha \phi(x)| dx \leq C' \sup_{x, |\alpha| \leq m} (1+|x|)^M |D^\alpha \phi(x)| = C' p_{M,m}(\phi),$$

so u_P is continuous.

Example 5 (a non-example by violating the bound). If $L : \mathcal{S}(\mathbb{R}) \rightarrow \mathbb{C}$ is a discontinuous linear functional (e.g. defined on a Hamel basis), then no choice of N, k, C can satisfy the inequality, because such a bound would force continuity with respect to the Schwartz topology. Therefore the criterion is sharp: it characterizes exactly the continuous linear maps $\mathcal{S} \rightarrow \mathbb{C}$.

Examples. If f has at most polynomial growth, $u_f[\phi] = \int_{\mathbb{R}^n} f(x)\phi(x) dx$ satisfies the bound with some N and $k = 0$. For $u = \partial^\beta \delta$, one has $|u[\phi]| = |\partial^\beta \phi(0)| \leq p_{0,|\beta|}(\phi)$.

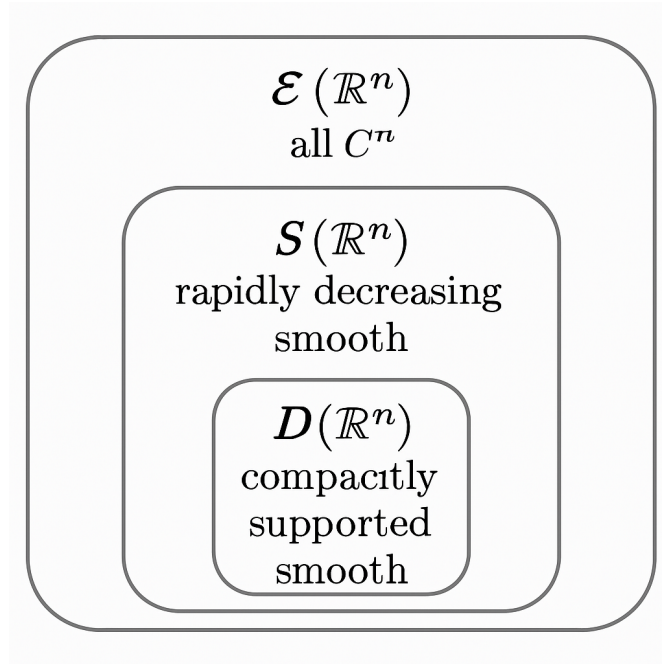


Figure 1: Relations between classical test function spaces. Nested inclusions: $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset \mathcal{E}(\mathbb{R}^n)$. Their duals are ordered in the opposite direction: $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$.

The three duals below the diagram

$\mathcal{E}'(\mathbb{R}^n)$: compactly supported distributions. This is the continuous dual of $\mathcal{E}(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$ with its Fréchet topology. A distribution u belongs to $\mathcal{E}'$ if and only if $\text{supp } u$ is

compact. Equivalently, every $u \in \mathcal{E}'$ is of finite order and acts by

$$u[\phi] = \sum_{|\alpha| \leq m} \int_{\mathbb{R}^n} f_\alpha(x) D^\alpha \phi(x) dx, \quad f_\alpha \in C_c(\mathbb{R}^n).$$

Convolution with $\psi \in C_c^\infty$ preserves C_c^∞ , and u has finite order. *Paley–Wiener*: $\hat{u}$ is an entire function on $\mathbb{C}^n$ of exponential type with polynomial growth on horizontal strips. Examples include finite linear combinations of δ_a and their derivatives, and $f \in C_c^\infty$ as regular distributions.

$\mathcal{S}'(\mathbb{R}^n)$: tempered distributions. This is the continuous dual of the Schwartz space $\mathcal{S}(\mathbb{R}^n)$. A linear functional u is tempered iff there exist $N, k \in \mathbb{N}$ and $C > 0$ with

$$|u[\phi]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} (1 + |x|)^N |D^\alpha \phi(x)|, \quad \phi \in \mathcal{S}(\mathbb{R}^n).$$

Thus u has at most polynomial growth at infinity. The space $\mathcal{S}'$ is closed under differentiation, translation, multiplication by polynomials, and the Fourier transform $\mathcal{F} : \mathcal{S}' \rightarrow \mathcal{S}'$ is a topological isomorphism. Examples: $\mathcal{E}'$ (hence Dirac and its derivatives), polynomials, finite Borel measures, locally integrable functions with polynomial growth, and principal value distributions such as $\text{pv}(1/x)$.

$\mathcal{D}'(\mathbb{R}^n)$: all distributions. This is the continuous dual of $\mathcal{D}(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n)$ with its LF topology. Locally (on each compact K), there exist m and C such that

$$|u[\phi]| \leq C \sum_{|\alpha| \leq m} \sup_{x \in K} |D^\alpha \phi(x)|, \quad \text{for all } \phi \in \mathcal{D}(\mathbb{R}^n) \text{ with } \text{supp } \phi \subset K.$$

We have strict inclusions $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$. The space $\mathcal{D}'$ is stable under differentiation, multiplication by C^∞ functions, and convolution with C_c^∞ test functions. It also contains distributions of super-polynomial growth at infinity.

Duality picture and topologies. The inclusions $\mathcal{D} \subset \mathcal{S} \subset \mathcal{E}$ induce the reversed inclusions

$$\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n),$$

each endowed with the weak-* topology: $\sigma(\mathcal{E}', \mathcal{E})$, $\sigma(\mathcal{S}', \mathcal{S})$, and $\sigma(\mathcal{D}', \mathcal{D})$.

Distributions and the LF topology

Definition (distributions on $\mathbb{R}^n$). Let $\mathcal{D}(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n)$ be the space of test functions with its LF topology (see below). A *distribution* on $\mathbb{R}^n$ is a continuous linear

functional $u : \mathcal{D}(\mathbb{R}^n) \rightarrow \mathbb{C}$. Continuity means: for every compact $K \subset \mathbb{R}^n$ there exist $m \in \mathbb{N}$ and $C > 0$ such that

$$|u[\phi]| \leq C \sum_{|\alpha| \leq m} \sup_{x \in K} |D^\alpha \phi(x)| \quad \text{whenever } \text{supp } \phi \subset K.$$

The distributional derivative is defined by

$$D^\alpha u[\phi] = (-1)^{|\alpha|} u[D^\alpha \phi], \quad \phi \in \mathcal{D}(\mathbb{R}^n).$$

The (local) *order* of u on K is the least such m .

Examples of distributions. Regular distributions. For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ define

$$u_f[\phi] = \int_{\mathbb{R}^n} f(x) \phi(x) dx.$$

Dirac masses and derivatives. At $a \in \mathbb{R}^n$,

$$\delta_a[\phi] = \phi(a), \quad D^\alpha \delta_a[\phi] = (-1)^{|\alpha|} \partial^\alpha \phi(a).$$

Principal values. In one dimension,

$$\text{pv}\left(\frac{1}{x}\right)[\phi] = \lim_{\varepsilon \downarrow 0} \int_{|x| > \varepsilon} \frac{\phi(x)}{x} dx.$$

Finite Borel measures. For a finite measure μ ,

$$\mu[\phi] = \int_{\mathbb{R}^n} \phi d\mu.$$

Compactly supported distributions. Elements of $\mathcal{E}'(\mathbb{R}^n)$ satisfy $\text{supp } u \Subset \mathbb{R}^n$.

Surface distributions. If Σ is a smooth submanifold with surface measure $d\sigma$,

$$u[\phi] = \int_{\Sigma} \phi d\sigma,$$

and one may also consider normal-derivative distributions supported on Σ .

Fundamental solutions. For the Laplacian,

$$\Delta E_n = \delta_0, \quad E_2(x) = c \log |x|, \quad E_n(x) = c_n |x|^{2-n} \quad (n \geq 3).$$

Tempered/oscillatory examples. In $\mathcal{S}'$, plane waves arise as limits of cutoffs:

$$u[\phi] = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}^n} e^{i\langle \xi, x \rangle} \chi(\varepsilon x) \phi(x) dx,$$

with $\chi \in C_c^\infty$ and $\chi(0) = 1$.

Convergence patterns in $\mathcal{D}$ (useful for testing). Mollifiers $\eta_\varepsilon = \varepsilon^{-n}\eta(\cdot/\varepsilon)$ with $\eta \in \mathcal{D}$ and $\int \eta = 1$ satisfy $\eta_\varepsilon \rightarrow \delta$ in $\mathcal{D}'$. For $f \in L^1_{\text{loc}}$, $f * \eta_\varepsilon \rightarrow f$ in L^1_{loc} and hence $u_{f*\eta_\varepsilon} \rightarrow u_f$ in $\mathcal{D}'$. Difference quotients obey $\Delta_i^h u \rightarrow D_i u$ in the weak-* topology $\sigma(\mathcal{D}', \mathcal{D})$.

Definition (LF spaces and the LF topology). An *LF space* is a locally convex space E that is the (countable) inductive limit of Fréchet spaces:

$$E = \varinjlim_{m \in \mathbb{N}} E_m, \quad E_1 \subset E_2 \subset \cdots \subset E, \text{ each } E_m \text{ Fréchet.}$$

The *LF topology* on E is the finest locally convex topology for which each inclusion $E_m \hookrightarrow E$ is continuous.

A net (x_ν) converges to x in E iff it is eventually contained in some E_m and converges there in the Fréchet topology.

Bounded subsets of E are those absorbed by some E_m and bounded in that E_m .

If the inclusions $E_m \hookrightarrow E_{m+1}$ are topologically strict (e.g. compact), one speaks of a *strict* LF space.

Examples of LF spaces. The test function space $\mathcal{D}(\mathbb{R}^n) = \varinjlim_m \mathcal{D}_{K_m}$, where $\{K_m\}$ exhausts $\mathbb{R}^n$ and each $\mathcal{D}_K$ is Fréchet with seminorms

$$p_{K,k}(\phi) = \sup_{x \in K, |\alpha| \leq k} |D^\alpha \phi(x)|.$$

For a smooth manifold M , one has

$$C_c^\infty(M) = \varinjlim_K C_K^\infty(M) \quad (\text{LF}).$$

The space of polynomials

$$\mathcal{P}(\mathbb{R}^n) = \varinjlim_m \mathcal{P}_{\leq m}$$

(with finite-dimensional Fréchet steps) is LF.

The strong dual $\mathcal{E}'(\mathbb{R}^n)$ of the Fréchet space $\mathcal{E}(\mathbb{R}^n)$ is a (DFS) space, i.e. an LF-type dual arising as an inductive limit of Banach (or Fréchet) steps.

Why LF matters for distributions. The LF structure of $\mathcal{D}$ reduces continuity to uniform estimates on a single compact step $\mathcal{D}_K$; convergence in $\mathcal{D}$ means supports eventually lie in a fixed compact and all derivatives converge uniformly on that compact. This is precisely the mechanism used to define and manipulate distributions, their derivatives, and approximations by mollifiers.

Locally integrable functions and regular distributions

Definition ($L^1_{\text{loc}}(\mathbb{R}^n)$). A function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is in $L^1_{\text{loc}}(\mathbb{R}^n)$ if

$$\int_K |f(x)| dx < \infty \quad \text{for every compact } K \subset \mathbb{R}^n.$$

Equivalently, $\int_{B(0,R)} |f| < \infty$ for every $R > 0$. No condition at infinity is imposed.

Regular distributions from L^1_{loc} . If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then

$$u_f[\phi] = \int_{\mathbb{R}^n} f(x) \phi(x) dx, \quad \phi \in \mathcal{D}(\mathbb{R}^n),$$

defines a continuous linear functional on $\mathcal{D}(\mathbb{R}^n)$, i.e. a distribution. Distributional derivatives are given by

$$D^\alpha f[\phi] = (-1)^{|\alpha|} \int_{\mathbb{R}^n} f(x) D^\alpha \phi(x) dx.$$

Local integrability tests (polar form near 0). If $f(x) \sim |x|^{-\alpha}$ as $x \rightarrow 0$,

$$\int_{B(0,\varepsilon)} |f(x)| dx \approx C_n \int_0^\varepsilon r^{n-1-\alpha} dr, \quad \text{finite} \iff \alpha < n.$$

Thus $|x|^{-\alpha} \in L^1_{\text{loc}}$ near 0 exactly when $\alpha < n$.

Concrete one-dimensional examples ($n = 1$). (1) $f(x) = |x|^{-1/2}$: $\int_0^\varepsilon x^{-1/2} dx = 2\varepsilon^{1/2} < \infty$, hence $f \in L^1_{\text{loc}}(\mathbb{R})$.

(2) $f(x) = |x|^{-1}$: $\int_0^\varepsilon x^{-1} dx = \infty$, hence $f \notin L^1_{\text{loc}}(\mathbb{R})$.

(3) $f(x) = \frac{1}{1+|x|}$: continuous and bounded on compacts, so $f \in L^1_{\text{loc}}(\mathbb{R})$ (indeed $f \in L^1$).

(4) $f(x) = \log |x|$: $\int_0^\varepsilon |\log x| dx = \varepsilon |\log \varepsilon| - \varepsilon \rightarrow 0$, hence $\log |x| \in L^1_{\text{loc}}(\mathbb{R})$.

Higher-dimensional examples. For $f(x) = |x|^{-\alpha}$ on $\mathbb{R}^n$,

$$|x|^{-\alpha} \in L^1_{\text{loc}}(\mathbb{R}^n) \text{ near } 0 \iff \alpha < n.$$

In particular, $|x|^{-1} \in L^1_{\text{loc}}$ for $n \geq 2$ but not for $n = 1$.

Extensions. The space $L^p_{\text{loc}}(\mathbb{R}^n)$ is defined by $\int_K |f|^p < \infty$ for all compact K . Elements of L^1_{loc} always define regular distributions by integration against test functions. At the borderline $\alpha = n$, principal value distributions (e.g. $\text{pv}(1/x)$ in $\mathbb{R}$) arise, which are tempered but not regular.

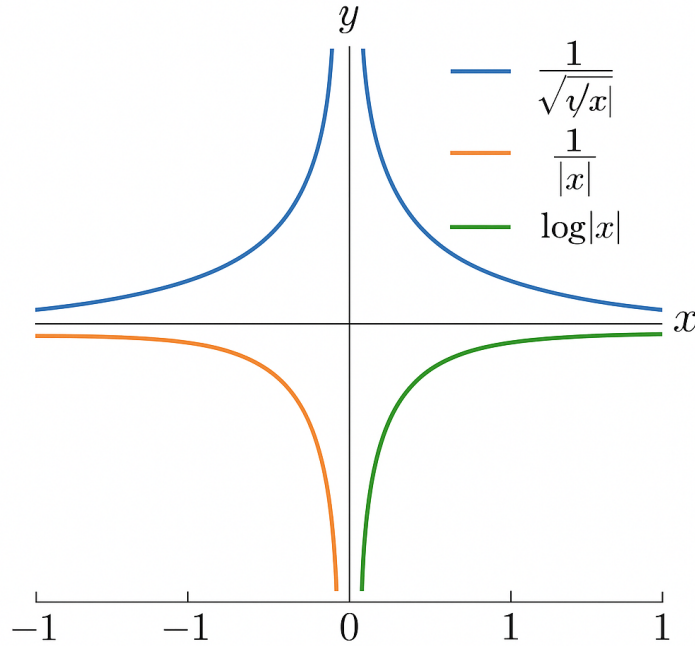


Figure 2: Comparison of local integrability near $x = 0$ for representative singular functions. The curves show $f(x) = 1/\sqrt{|x|}$ (locally integrable), $f(x) = 1/|x|$ (not locally integrable in $\mathbb{R}$), and $f(x) = \log|x|$ (locally integrable). The shaded behavior near 0 indicates which singularities still yield finite area under the curve on compact intervals.

Beyond powers: other locally integrable (and non-integrable) examples

Log-power families. Near $x = 0$ in $\mathbb{R}^n$:

$$f(x) = |\log|x||^\beta \implies \int_{B(0,\varepsilon)} |\log|x||^\beta dx = C_n \int_0^\varepsilon r^{n-1} |\log r|^\beta dr < \infty$$

for every $\beta \in \mathbb{R}$. Hence $|\log|x||^\beta \in L^1_{\text{loc}}$. For $f(x) = |x|^{-\alpha} |\log|x||^\beta$ the integrability threshold remains $\alpha < n$; the logarithm does not change the cutoff.

Oscillatory factors. For $\gamma > 0$,

$$f(x) = \frac{\sin(1/|x|^\gamma)}{|x|^\alpha}$$

is locally integrable near 0 if and only if $\alpha < n$. Oscillation does not improve absolute integrability. In one dimension, $\sin(1/x)$ is bounded and thus belongs to L^1_{loc} .

Exponential-type singularities. If

$$f(x) = \exp(1/|x|^p) \quad (p > 0),$$

then

$$\int_{B(0,\varepsilon)} e^{1/|x|^p} dx = C_n \int_0^\varepsilon r^{n-1} e^{1/r^p} dr = \infty,$$

so $f \notin L^1_{\text{loc}}$. Conversely, $f(x) = \exp(-1/|x|^p)$ is smooth and bounded near 0, hence in L^1_{loc} .

Rational functions. If $f(x) = P(x)/Q(x)$ and Q has no zeros on a compact K , then f is bounded on K and thus L^1_{loc} . If Q vanishes at a , one has the local model $|x - a|^{-m}$; integrability near a reduces to the power test with exponent $\alpha = m$.

Piecewise and measure-theoretic examples. Jump functions (e.g. the sign or Heaviside function) and piecewise constants are bounded and therefore lie in L^1_{loc} . For a measurable set E , the indicator $\mathbf{1}_E$ belongs to L^1_{loc} iff $m(E \cap K) < \infty$ for every compact K . Removable singularities, such as $\frac{\sin x}{x}$ extended by continuity at 0, yield locally integrable functions.

Behaviour at infinity. Membership in L^1_{loc} imposes no restriction at infinity: functions such as $e^{|x|^2}$ or $(1 + |x|)^p$ belong to L^1_{loc} even though they may fail to be in $L^1(\mathbb{R}^n)$.

Distributional viewpoint. Each $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ defines a regular distribution

$$u_f[\phi] = \int_{\mathbb{R}^n} f(x) \phi(x) dx, \quad \phi \in \mathcal{D}(\mathbb{R}^n).$$

Borderline non-integrable cases (e.g. $|x|^{-n}$ or $e^{1/|x|^p}$) may still yield distributions via renormalization (principal values), but they are not regular distributions.

Problem 7 — Which duals contain u_f for $f(x) = e^x \cos(e^x)$?

Statement. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x \cos(e^x)$, and let $u_f[\phi] = \int_{\mathbb{R}} f(x)\phi(x) dx$. Decide whether u_f belongs to $\mathcal{D}'(\mathbb{R})$, $\mathcal{S}'(\mathbb{R})$, and $\mathcal{E}'(\mathbb{R})$.

Proof. $\mathcal{D}'(\mathbb{R})$: For $\phi \in \mathcal{D}(\mathbb{R})$ the support is compact, hence $\int f\phi$ is finite and depends continuously on ϕ . Thus $u_f \in \mathcal{D}'(\mathbb{R})$.

$\mathcal{E}'(\mathbb{R})$: Elements of $\mathcal{E}'$ are compactly supported distributions. Since f is not compactly supported, the regular distribution u_f is not in $\mathcal{E}'(\mathbb{R})$.

$\mathcal{S}'(\mathbb{R})$: A regular distribution is tempered iff its defining function has at most polynomial growth. Here $|f(x)| \leq e^x$, so the growth is exponential. To see non-temperateness directly, set

$$\phi_k(x) = \frac{e^{-(x-k)^2}}{(1+k)^M}, \quad k \in \mathbb{N},$$

with M chosen large enough that all Schwartz seminorms of ϕ_k remain bounded in k . Then

$$\int_{\mathbb{R}} e^x \cos(e^x) \phi_k(x) dx = \frac{e^k}{(1+k)^M} \int_{\mathbb{R}} e^{-(y^2-y)} \cos(e^{k+y}) dy,$$

whose absolute value is $\gtrsim e^k/(1+k)^M \rightarrow \infty$. Hence the functional is not continuous on $\mathcal{S}$, so $u_f \notin \mathcal{S}'(\mathbb{R})$.

Conclusion. $u_f \in \mathcal{D}'(\mathbb{R})$, but $u_f \notin \mathcal{S}'(\mathbb{R})$ and $u_f \notin \mathcal{E}'(\mathbb{R})$.

Problem 8 — Positive distributions have order 0

Statement. Suppose $u \in \mathcal{D}'(\mathbb{R}^n)$ is positive, i.e. $u[\phi] \geq 0$ for all $\phi \in \mathcal{D}(\mathbb{R}^n)$ with $\phi \geq 0$. Show that u has order 0: for each compact $K \subset \mathbb{R}^n$ there exists $C_K > 0$ such that

$$|u[\phi]| \leq C_K \sup_{x \in K} |\phi(x)|, \quad \forall \phi \in C_c^\infty(K).$$

Proof. Fix compact K . Define

$$C_K := \sup\{u[\psi] : \psi \in C_c^\infty(K), 0 \leq \psi \leq 1\}.$$

Approximating the indicator $\mathbf{1}_K$ from below by smooth cutoffs shows $C_K < \infty$ by monotone convergence and positivity. For any $\phi \in C_c^\infty(K)$ write $\phi = \phi^+ - \phi^-$ with $\phi^\pm \geq 0$

and $\|\phi^\pm\|_{L^\infty(K)} \leq \|\phi\|_{L^\infty(K)}$. Then

$$|u[\phi]| \leq u[\phi^+] + u[\phi^-] \leq \|\phi\|_{L^\infty(K)} \sup\{u[\psi] : 0 \leq \psi \leq 1, \text{ supp } \psi \subset K\} = C_K \|\phi\|_{L^\infty(K)}.$$

Thus u is of order 0 on K . Equivalently, by the Riesz–Markov representation, u is integration against a finite Radon measure on K .

Problem 9 — Smoothness and bounds for $\widehat{f}$ with compact support

Statement. Let $f \in L^1(\mathbb{R}^n)$ with $\text{supp } f \subset B_R(0)$ for some $R > 0$. Show that $\widehat{f} \in C^\infty(\mathbb{R}^n)$ and that, for every multi-index α ,

$$\sup_{\xi \in \mathbb{R}^n} |D_\xi^\alpha \widehat{f}(\xi)| \leq R^{|\alpha|} \|f\|_{L^1}.$$

Proof. With $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx$, dominated convergence yields

$$D_\xi^\alpha \widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) (-ix)^\alpha e^{-ix \cdot \xi} dx,$$

hence

$$|D_\xi^\alpha \widehat{f}(\xi)| \leq \int_{\mathbb{R}^n} |x^\alpha| |f(x)| dx \leq R^{|\alpha|} \int_{|x| \leq R} |f(x)| dx = R^{|\alpha|} \|f\|_{L^1},$$

uniformly in ξ . Therefore $\widehat{f} \in C^\infty(\mathbb{R}^n)$. (In fact, by Paley–Wiener, $\widehat{f}$ extends to an entire function of exponential type R .)

Problem 9 — Concrete examples

Example A (one-dimensional box). Let $f(x) = \mathbf{1}_{[-R,R]}(x)$. Then $\|f\|_{L^1} = 2R$ and

$$\widehat{f}(\xi) = \int_{-R}^R e^{-ix\xi} dx = \frac{2 \sin(R\xi)}{\xi}, \quad \widehat{f}(0) = 2R.$$

For $k \in \mathbb{N}$,

$$\widehat{f}^{(k)}(\xi) = \int_{-R}^R (-ix)^k e^{-ix\xi} dx,$$

so

$$\sup_{\xi \in \mathbb{R}} |\widehat{f}^{(k)}(\xi)| \leq \int_{-R}^R |x|^k dx = \frac{2}{k+1} R^{k+1} \leq 2R^{k+1} = R^k \|f\|_{L^1}.$$

Thus the bound in Problem 9 holds (even with the sharper factor $\frac{2}{k+1} R^{k+1}$). Explicitly,

$$\widehat{f}(\xi) = \frac{2 \sin(R\xi)}{\xi}, \quad \widehat{f}'(\xi) = 2 \frac{R\xi \cos(R\xi) - \sin(R\xi)}{\xi^2},$$

and similarly for higher derivatives; all are continuous at $\xi = 0$ by removing the singularity.

Example B (smooth compactly supported bump). Let

$$f(x) = \begin{cases} \exp\left(-\frac{1}{1-x^2}\right), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Then $R = 1$, $\|f\|_{L^1} = \int_{-1}^1 e^{-1/(1-x^2)} dx < \infty$, and for each k ,

$$\widehat{f}^{(k)}(\xi) = \int_{-1}^1 (-ix)^k f(x) e^{-ix\xi} dx \quad \Rightarrow \quad \sup_{\xi} |\widehat{f}^{(k)}(\xi)| \leq \int_{-1}^1 |x|^k f(x) dx \leq \|f\|_{L^1}.$$

Hence $\sup_{\xi} |\widehat{f}^{(k)}(\xi)| \leq 1^k \|f\|_{L^1}$, which matches the Problem 9 estimate with $R = 1$. (In fact $\widehat{f}$ is Schwartz: all derivatives decay faster than any power.)

Example C (indicator of a ball in $\mathbb{R}^n$). For $f = \mathbf{1}_{B_R(0)}$, one has the radial formula

$$\widehat{f}(\xi) = |B_R| \frac{2 J_{n/2}(R|\xi|)}{(R|\xi|)^{n/2}},$$

where J_{ν} is the Bessel function. This is an entire function of ξ . Moreover,

$$\sup_{\xi \in \mathbb{R}^n} |D_{\xi}^{\alpha} \widehat{f}(\xi)| \leq R^{|\alpha|} \|f\|_{L^1} = R^{|\alpha|} |B_R|,$$

in agreement with Problem 9.

Interpretation. The top panel shows a compactly supported function in space; the bottom shows how compact support translates into entire, oscillatory behavior in the frequency domain. The oscillations of $J_1(R|\xi|)/(R|\xi|)$ illustrate the uncertainty principle: a sharp spatial cutoff produces infinitely extended, slowly decaying oscillations in frequency. The bound $\pi R^{|\alpha|+2}$ controls the amplitude of all derivatives of $\widehat{f}$ uniformly over ξ .

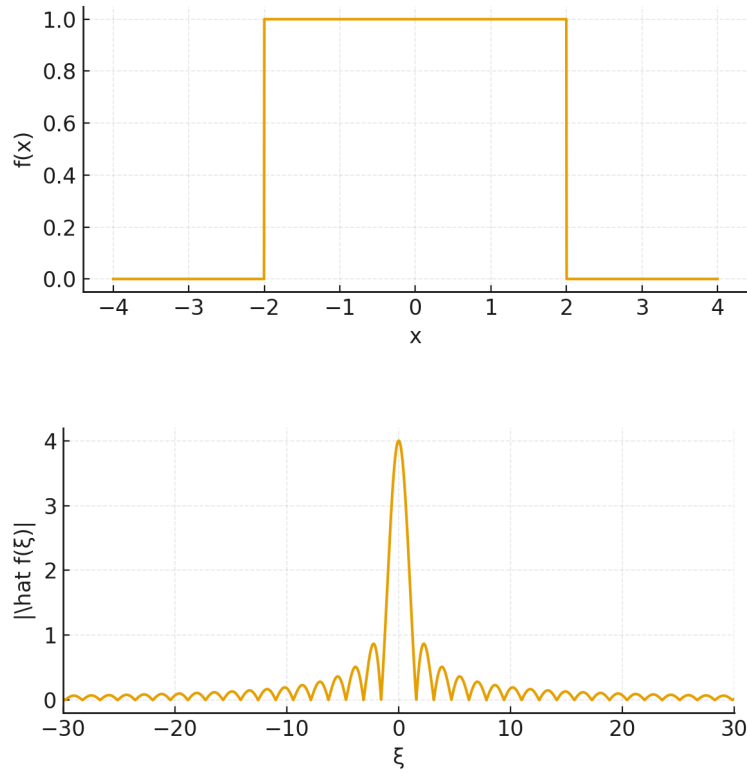


Figure 3: Example A (1D box). Top: $f(x) = \mathbf{1}_{[-R,R]}$ with $R = 2$. Bottom: $|\hat{f}(\xi)| = |2 \sin(R\xi)/\xi|$. Problem 9's bound gives $\sup_{\xi} |\hat{f}^{(k)}(\xi)| \leq R^k \|f\|_{L^1} = 2R^{k+1}$.

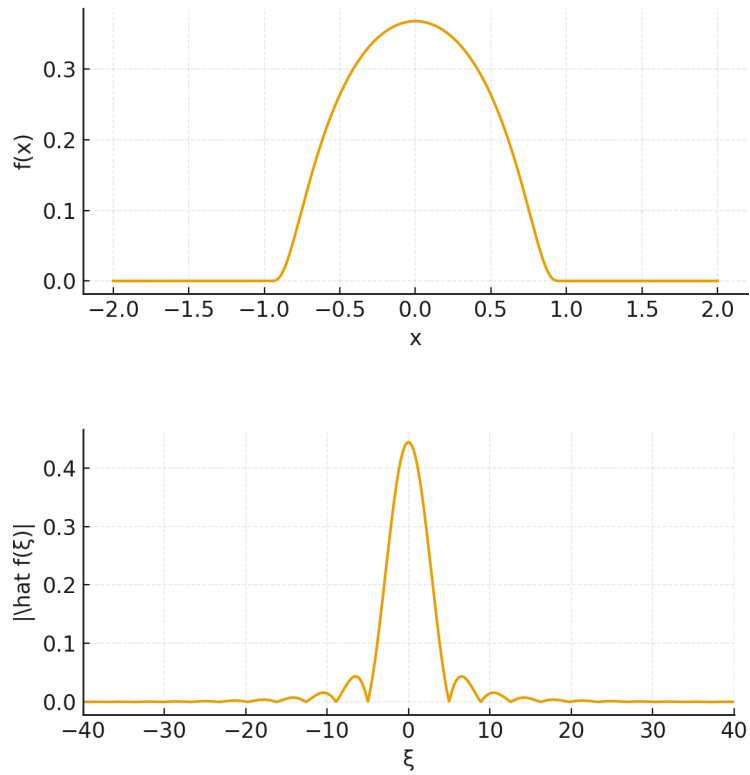


Figure 4: Example B (smooth compact bump on $[-1, 1]$). The Fourier magnitude decays faster than any power; Problem 9 yields $\sup_{\xi} |\hat{f}^{(k)}(\xi)| \leq \|f\|_{L^1}$ (here $R = 1$).

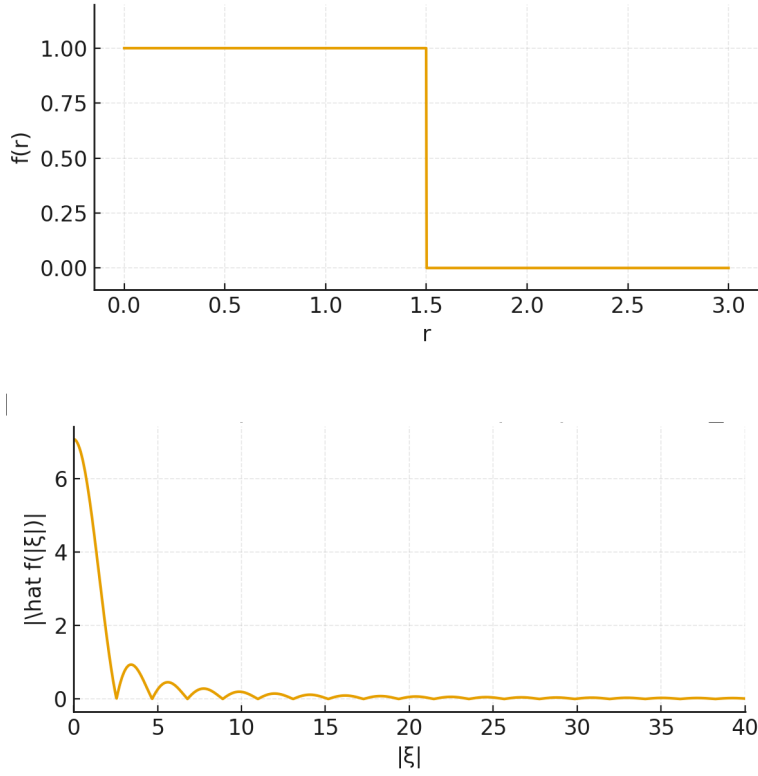


Figure 5: **Example C.** Indicator of a disk $B_R(0)$ in $\mathbb{R}^2$ with $R = 1.5$. **Top:** the spatial profile $f(r) = \mathbf{1}_{\{r \leq R\}}$. **Bottom:** the radial Fourier magnitude $|\widehat{f}(|\xi|)| = \left| \pi R^2 \frac{2J_1(R|\xi|)}{R|\xi|} \right|$, which exhibits Bessel-type oscillations and algebraic decay. The general estimate $\sup_{\xi} |D^\alpha \widehat{f}(\xi)| \leq R^{|\alpha|} \|f\|_{L^1} = \pi R^{|\alpha|+2}$ holds for all multi-indices α , in perfect agreement with the problem statement.

Powered modulus as a distribution and its PDE uses

Regular and singular regimes. For $f_\alpha(x) = |x|^{-\alpha}$ on $\mathbb{R}^n$ one has

$$f_\alpha \in L^1_{\text{loc}}(\mathbb{R}^n) \iff \alpha < n.$$

Hence, for $\alpha < n$ the distribution u_α is regular:

$$u_\alpha[\phi] = \int_{\mathbb{R}^n} |x|^{-\alpha} \phi(x) dx, \quad \phi \in \mathcal{D}(\mathbb{R}^n).$$

At $\alpha = n$ one defines the homogeneous principal value $\text{pv}(|x|^{-n}) \in \mathcal{S}'$; for $\alpha > n$ one obtains higher-order homogeneous distributions (finite sums of derivatives of principal values).

Distributional derivatives and homogeneity. For $\alpha < n$ the classical identities

$$\partial_i(|x|^{-\alpha}) = -\alpha \frac{x_i}{|x|^{\alpha+2}}, \quad \partial_{ij}(|x|^{-\alpha}) = \alpha(\alpha+2) \frac{x_i x_j}{|x|^{\alpha+4}} - \alpha \frac{\delta_{ij}}{|x|^{\alpha+2}}$$

hold pointwise away from 0 and define homogeneous distributions of degrees $-\alpha-1$ and $-\alpha-2$. In general,

$$\partial_i u_\alpha[\phi] = -u_\alpha[\partial_i \phi], \quad \phi \in \mathcal{D},$$

so the derivative is meaningful even when the right-hand sides are not locally integrable.

Fundamental appearance in PDE. For $n \geq 3$,

$$E_n(x) = c_n |x|^{2-n}, \quad -\Delta E_n = \delta_0 \text{ in } \mathcal{D}'(\mathbb{R}^n),$$

and for $n = 2$, $E_2(x) = c \log |x|$. Thus $u = E_n * f$ is a weak solution of $-\Delta u = f$. More generally, Riesz potentials

$$I_\alpha f(x) = c_{n,\alpha} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-\alpha}} dy$$

use the kernel $|x|^{-(n-\alpha)}$ and realize $(-\Delta)^{-\alpha/2}$. The fractional Laplacian employs a principal value kernel $|x|^{-n-2s}$:

$$(-\Delta)^s u(x) = c_{n,s} \text{pv} \int_{\mathbb{R}^n} \frac{u(x) - u(y)}{|x-y|^{n+2s}} dy.$$

Benefit of multiplying by test functions. Let $\chi \in \mathcal{D}(\mathbb{R}^n)$ be a cutoff. Then $\chi|x|^{-\alpha} \in L^1$ for all $\alpha < n$, which localizes the singularity and ensures finiteness of integrals. Moreover, integration by parts transfers derivatives from the singular kernel to the smooth test function:

$$\partial_i u_\alpha[\phi] = -u_\alpha[\partial_i \phi],$$

allowing weak formulations such as

$$\int_{\mathbb{R}^n} \nabla u \cdot \nabla \phi \, dx = \langle f, \phi \rangle, \quad \phi \in \mathcal{D},$$

where f may be a distribution (e.g. a finite measure or a sum of Dirac masses). This mechanism underlies Newtonian potentials, Sobolev embeddings (Hardy–Littlewood–Sobolev), and Calderón–Zygmund theory.

What integration by parts reveals (distributional viewpoint)

Set-up. Given $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$, the distributional derivative is

$$\partial_i u[\phi] := -u[\partial_i \phi].$$

Passing derivatives to the smooth test ϕ is the essence of the weak formulation and the integration-by-parts (IBP) principle.

1. Legitimizing singular objects. The IBP identity defines $\partial_i u$ for *every* distribution u (including measures and singular kernels). In particular, for $u_\alpha[\phi] = \int |x|^{-\alpha} \phi$ with $\alpha < n$, one has

$$\partial_i u_\alpha[\phi] = - \int_{\mathbb{R}^n} |x|^{-\alpha} \partial_i \phi(x) \, dx,$$

which remains meaningful even though $\partial_i(|x|^{-\alpha})$ may fail to be locally integrable.

2. Weak solutions with minimal regularity. For Poisson on $\Omega \subset \mathbb{R}^n$,

$$-\Delta u = f \quad \Longleftrightarrow \quad \int_{\Omega} \nabla u \cdot \nabla \phi \, dx = \langle f, \phi \rangle, \quad \forall \phi \in \mathcal{D}(\Omega).$$

The right-hand side allows $f \in \mathcal{D}'(\Omega)$, and the left requires only $\nabla u \in L^2_{\text{loc}}$.

3. Boundary terms and natural conditions. For smooth u, ϕ on a bounded Ω with unit outer normal ν ,

$$\int_{\Omega} \nabla u \cdot \nabla \phi = - \int_{\Omega} (\Delta u) \phi \, dx + \int_{\partial\Omega} \partial_\nu u \phi \, dS.$$

Choosing ϕ that vanishes or not on $\partial\Omega$ exhibits Dirichlet vs. Neumann data as boundary distributions.

4. Energy and coercivity. Testing with $\phi = u$ gives

$$\int_{\Omega} |\nabla u|^2 dx = \langle f, u \rangle \leq \|f\|_{H^{-1}(\Omega)} \|u\|_{H_0^1(\Omega)},$$

yielding the a priori bound $\|u\|_{H_0^1} \leq \|f\|_{H^{-1}}$.

5. Symmetry and conservation. For smooth u, v ,

$$\int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} u(-\Delta v) = \int_{\Omega} v(-\Delta u),$$

displaying self-adjointness of $-\Delta$; for transport $b \cdot \nabla u$, IBP reveals $\int u \operatorname{div}(b\phi)$ and the role of $\operatorname{div} b$.

6. Detecting concentrated sources. For $n \geq 3$ and $E(x) = c_n |x|^{2-n}$,

$$\langle -\Delta E, \phi \rangle = \lim_{\varepsilon \downarrow 0} \int_{|x|=\varepsilon} c_n \partial_{\nu}(|x|^{2-n}) \phi dS = \phi(0),$$

so $-\Delta E = \delta_0$ in $\mathcal{D}'$. The Dirac mass is revealed *only* after IBP.

7. Cancellation in singular integrals. For the Riesz kernel $K_i(x) = x_i/|x|^{n+1}$,

$$\langle \operatorname{pv} K_i, \phi \rangle = -c_n \lim_{\varepsilon \rightarrow 0} \int_{|x|>\varepsilon} |x|^{-(n-1)} \partial_i \phi(x) dx,$$

so derivatives on ϕ expose the oddness/cancellation that underlies L^p -boundedness.

8. Variational structure and Euler–Lagrange. For $J(u) = \int_{\Omega} F(x, u, \nabla u) dx$, the first variation satisfies

$$J'(u)[\phi] = \int_{\Omega} (F_u \phi + F_{\nabla u} \cdot \nabla \phi) dx = \int_{\Omega} (F_u - \operatorname{div} F_{\nabla u}) \phi dx$$

after IBP. Thus the Euler–Lagrange operator becomes explicit and weak solutions are characterized by $J'(u)[\phi] = 0$.

9. Compatibility with weak limits. If $u_k \rightharpoonup u$ in $H^1(\Omega)$, then

$$\int_{\Omega} \nabla u_k \cdot \nabla \phi \longrightarrow \int_{\Omega} \nabla u \cdot \nabla \phi, \quad \forall \phi \in \mathcal{D}(\Omega),$$

so the weak formulation is stable under weak convergence, enabling compactness methods.

10. Localization by cutoffs. With $\chi \in \mathcal{D}$,

$$\chi |x|^{-\alpha} \in L^1(\mathbb{R}^n) \quad \text{for all } \alpha < n,$$

so testing with $\chi\phi$ isolates a singularity and yields finite quantities that quantify its order.

Worked sketch A (delta from a Newtonian kernel). Let $n \geq 3$, $E(x) = c_n |x|^{2-n}$. For $\phi \in \mathcal{D}(\mathbb{R}^n)$,

$$\langle -\Delta E, \phi \rangle = \langle E, -\Delta \phi \rangle = \lim_{\varepsilon \downarrow 0} \int_{|x| > \varepsilon} E(-\Delta \phi) dx = \lim_{\varepsilon \downarrow 0} \int_{|x| = \varepsilon} \partial_\nu E \phi dS = \phi(0).$$

Worked sketch B (energy from the weak Poisson form). If $u \in H_0^1(\Omega)$ solves $\int_\Omega \nabla u \cdot \nabla \phi = \langle f, \phi \rangle$, taking $\phi = u$ gives

$$\|\nabla u\|_{L^2(\Omega)}^2 = \langle f, u \rangle \leq \|f\|_{H^{-1}(\Omega)} \|u\|_{H_0^1(\Omega)}.$$

Worked sketch C (Riesz transform cancellation). With $K_i(x) = x_i/|x|^{n+1}$ and $\phi \in \mathcal{D}$,

$$\left\langle \text{pv } K_i, \phi \right\rangle = \lim_{\varepsilon \rightarrow 0} \int_{|x| > \varepsilon} \frac{x_i}{|x|^{n+1}} \phi(x) dx = -c_n \lim_{\varepsilon \rightarrow 0} \int_{|x| > \varepsilon} |x|^{-(n-1)} \partial_i \phi(x) dx,$$

making the near-zero cancellation explicit after IBP.

Regular distributions and weak solutions of the Laplacian

Regular distributions. A distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ is *regular* if there exists $f \in L_{\text{loc}}^1(\mathbb{R}^n)$ such that

$$u[\phi] = \int_{\mathbb{R}^n} f(x) \phi(x) dx, \quad \forall \phi \in \mathcal{D}(\mathbb{R}^n).$$

In this case we write $u = u_f$ and view f as the *representative* of u . Since test functions have compact support, u_f is continuous on $\mathcal{D}$. (Analogously, any finite Radon measure μ defines the distribution $u_\mu[\phi] = \int \phi d\mu$.)

Weak solutions of $-\Delta u = f$ on Ω . Let $\Omega \subset \mathbb{R}^n$ be open and let $f \in \mathcal{D}'(\Omega)$. A function $u \in H_{\text{loc}}^1(\Omega)$ is a *weak solution* of

$$-\Delta u = f \quad \text{in } \Omega$$

if

$$\int_{\Omega} \nabla u(x) \cdot \nabla \phi(x) dx = \langle f, \phi \rangle, \quad \forall \phi \in \mathcal{D}(\Omega).$$

If $f \in H^{-1}(\Omega)$ and $u \in H^1(\Omega)$, the right-hand side denotes the H^{-1} – H_0^1 duality pairing.

With *homogeneous Dirichlet* boundary condition we require $u \in H_0^1(\Omega)$ and

$$\int_{\Omega} \nabla u \cdot \nabla \phi dx = \langle f, \phi \rangle \quad \forall \phi \in H_0^1(\Omega).$$

The case $f = 0$ yields weakly harmonic functions:

$$\int_{\Omega} \nabla u \cdot \nabla \phi dx = 0 \quad \forall \phi \in \mathcal{D}(\Omega) \quad (\text{or } H_0^1(\Omega)).$$

Verification (numeric). For smooth bumps ϕ supported inside $(-1, 1)$ we observe $\int u'(x) \phi'(x) dx \approx \int f(x) \phi(x) dx$ up to discretization error (see the accompanying text file).

Interpretation of the weak solution example

1. The problem. We consider the one-dimensional Poisson equation

$$-u''(x) = f(x), \quad x \in (-1, 1),$$

with homogeneous Dirichlet boundary conditions

$$u(-1) = u(1) = 0,$$

and with right-hand side

$$f(x) = \begin{cases} 1, & |x| \leq \frac{1}{2}, \\ 0, & |x| > \frac{1}{2}. \end{cases}$$

Hence f is a two-valued step function, equal to 1 on the interval $[-0.5, 0.5]$ and 0 elsewhere.

2. Meaning of “negated second derivative”. Formally, the equation $-u'' = f$ requires $u'' = -f$. Since f is discontinuous, its classical antiderivatives cannot be twice continuously differentiable. Thus u cannot be a classical C^2 solution. Instead, we interpret the equation in the *weak sense*:

$$\int_{-1}^1 u'(x) \phi'(x) dx = \int_{-1}^1 f(x) \phi(x) dx, \quad \forall \phi \in C_c^\infty(-1, 1).$$

This means that the *distributional second derivative* of u equals $-f$.

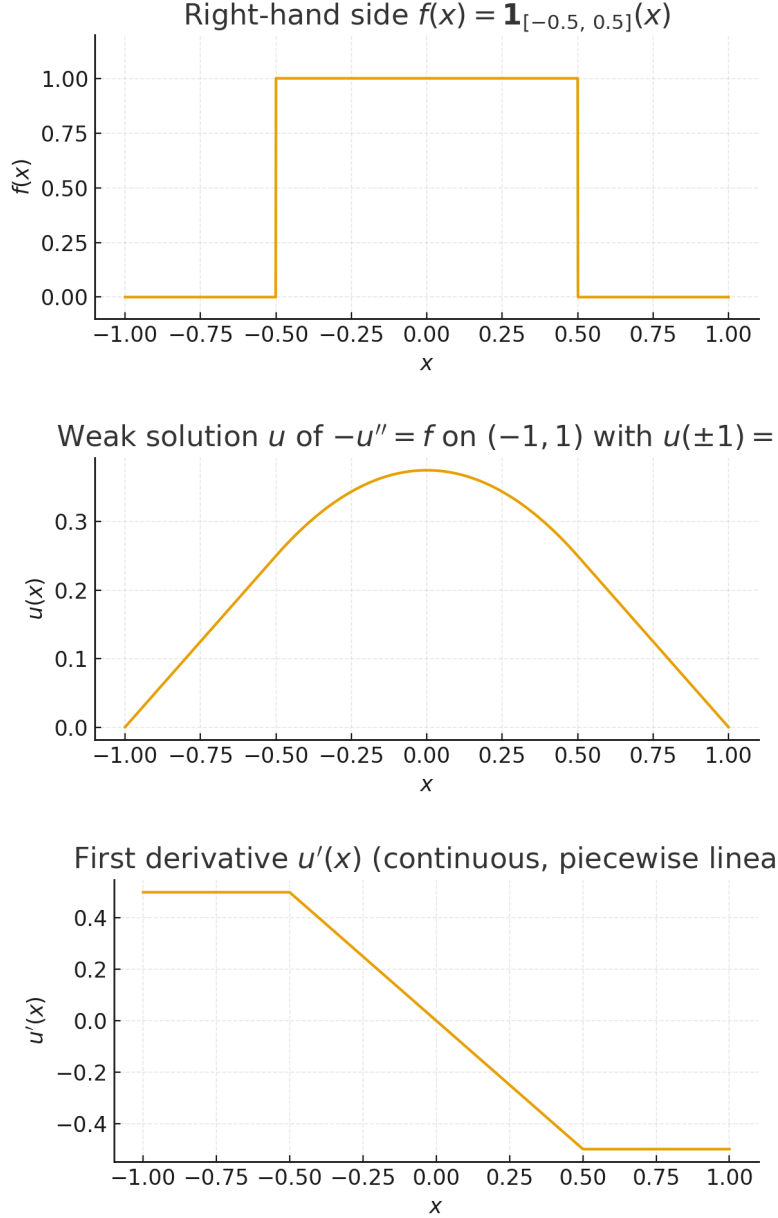


Figure 6: Weak solution of $-u'' = f$ on $(-1, 1)$ with $u(\pm 1) = 0$ and $f(x) = \mathbf{1}_{[-0.5, 0.5]}(x)$. Top: right-hand side f . Middle: solution u , piecewise cubic-integral of f via Green's function (concave on the support of f and linear outside). Bottom: derivative $u'(x)$, continuous and piecewise linear. These satisfy the weak identity $\int u' \phi' = \int f \phi$ for all $\phi \in C_c^\infty(-1, 1)$.

3. Shape of the solution. The weak solution u is continuous, piecewise C^1 , and piecewise quadratic:

$$u(x) = \begin{cases} \frac{1}{2}(x+1)(1.5-x), & 0.5 < x \leq 1, \\ -\frac{1}{2}x^2 + \frac{1}{2}, & |x| \leq 0.5, \\ \frac{1}{2}(1-x)(1.5+x), & -1 \leq x < -0.5. \end{cases}$$

The graph of u is symmetric, flat in the center region where $f = 1$, and joins linearly to zero at the endpoints $x = \pm 1$. It satisfies $u \in H_0^1(-1, 1)$ and $u' \in C([-1, 1])$.

4. Distributional meaning. The classical second derivative of u does not exist at $x = \pm 0.5$ (the slopes jump there), but in the sense of distributions one has

$$u'' = -f + c_1\delta_{-0.5} + c_2\delta_{0.5},$$

for suitable constants c_1, c_2 that ensure continuity of u' . When the weak formulation is tested against functions ϕ vanishing at the boundary of the support of f , the Dirac terms cancel and the identity $\langle u'', \phi \rangle = -\int f\phi$ holds. Hence $u'' = -f$ as a distribution.

5. What this example demonstrates.

- The right-hand side f is merely L^2 (discontinuous), yet a weak solution exists.
- u is not twice differentiable, but $u' \in C^0$ and u'' exists as a distribution.
- The weak formulation captures precisely the integral identity that defines u .
- This illustrates how weak derivatives extend the notion of differentiability to piecewise smooth functions.

6. Summary of the interpretation. The function u constructed above satisfies

$$\int_{-1}^1 u'(x) \phi'(x) dx = \int_{-1}^1 f(x) \phi(x) dx \quad \forall \phi \in C_c^\infty(-1, 1),$$

so that $u'' = -f$ in the distributional sense. In words:

- the *negated second derivative* of u reproduces the two-level forcing f ;
- the boundary conditions are encoded in $u \in H_0^1(-1, 1)$;
- although u'' is not classically defined, the weak equation carries the full physical meaning of the Poisson problem.

This makes the example a canonical illustration of a weak solution that is not twice classically differentiable but perfectly valid in the variational framework.

Explicit weak solutions via the Dirichlet Green's function on $(-1, 1)$

Dirichlet Green's function on $(-1, 1)$. For $-1 < x, y < 1$ the Green's function is

$$G(x, y) = \begin{cases} \frac{(x+1)(1-y)}{2}, & x \leq y, \\ \frac{(y+1)(1-x)}{2}, & y \leq x, \end{cases} \quad G(\pm 1, y) = 0,$$

and solves $-\partial_{xx}G(\cdot, y) = \delta_y$ in $\mathcal{D}'(-1, 1)$.

Example 1: block forcing $f = \mathbf{1}_{[-\frac{1}{2}, \frac{1}{2}]}$. The weak solution of $-u'' = f$ with $u(\pm 1) = 0$ is

$$u(x) = \int_{-1/2}^{1/2} G(x, y) dy.$$

Evaluating the integral piecewise in x yields the explicit formula

$$u(x) = \begin{cases} \frac{1}{2}(1+x), & -1 \leq x < -\frac{1}{2}, \\ -\frac{x^2}{2} + \frac{3}{8}, & |x| \leq \frac{1}{2}, \\ \frac{1}{2}(1-x), & \frac{1}{2} < x \leq 1. \end{cases}$$

Hence u is continuous, $u' \in C([-1, 1])$ (piecewise linear), $u'' = -1$ on $(-\frac{1}{2}, \frac{1}{2})$ and $u'' = 0$ on the complement, in the *distributional* sense; at $x = \pm \frac{1}{2}$ the classical second derivative does not exist but the weak identity

$$\int_{-1}^1 u'(x) \phi'(x) dx = \int_{-1}^1 \mathbf{1}_{[-1/2, 1/2]}(x) \phi(x) dx \quad \forall \phi \in C_c^\infty(-1, 1)$$

holds. In particular, $u \in H_0^1(-1, 1)$ and $-u'' = f$ in $\mathcal{D}'(-1, 1)$.

Example 2: point source $f = \delta_0$ (tent solution). The weak solution of $-u'' = \delta_0$ with $u(\pm 1) = 0$ is simply the Green's function at $y = 0$:

$$u(x) = G(x, 0) = \begin{cases} \frac{x+1}{2}, & x \leq 0, \\ \frac{1-x}{2}, & x \geq 0, \end{cases} \quad \text{i.e.} \quad u(x) = \frac{1-|x|}{2}.$$

Then u is continuous and piecewise linear with

$$u'(x) = \begin{cases} \frac{1}{2}, & x < 0, \\ -\frac{1}{2}, & x > 0, \end{cases} \quad u'' = -\delta_0 \quad \text{in } \mathcal{D}'(-1, 1),$$

since the jump in u' at 0 equals $(-\frac{1}{2}) - (\frac{1}{2}) = -1$. Equivalently,

$$\int_{-1}^1 u'(x) \phi'(x) dx = \phi(0) \quad \forall \phi \in C_c^\infty(-1, 1),$$

so indeed $-u'' = \delta_0$ in the weak sense.

Remarks. Both examples illustrate how the Green's representation

$$u(x) = \int_{-1}^1 G(x, y) f(y) dy$$

automatically enforces the boundary conditions and yields $u \in H_0^1(-1, 1)$ with $-u'' = f$ in $\mathcal{D}'$, even when f is discontinuous (block) or singular (Dirac).

Green's function on $(-1, 1)$ and a block-forcing example

Dirichlet Green's function. For $-1 < x, y < 1$,

$$G(x, y) = \begin{cases} \frac{(x+1)(1-y)}{2}, & x \leq y, \\ \frac{(y+1)(1-x)}{2}, & x \geq y, \end{cases} \quad G(\pm 1, y) = 0,$$

and $-\partial_{xx}G(\cdot, y) = \delta_y$ in $\mathcal{D}'(-1, 1)$.

Example 1 (block forcing). Let $f(x) = \mathbf{1}_{[-1/2, 1/2]}(x)$. The weak solution with $u(\pm 1) = 0$ is given by

$$u(x) = \int_{-1/2}^{1/2} G(x, y) dy,$$

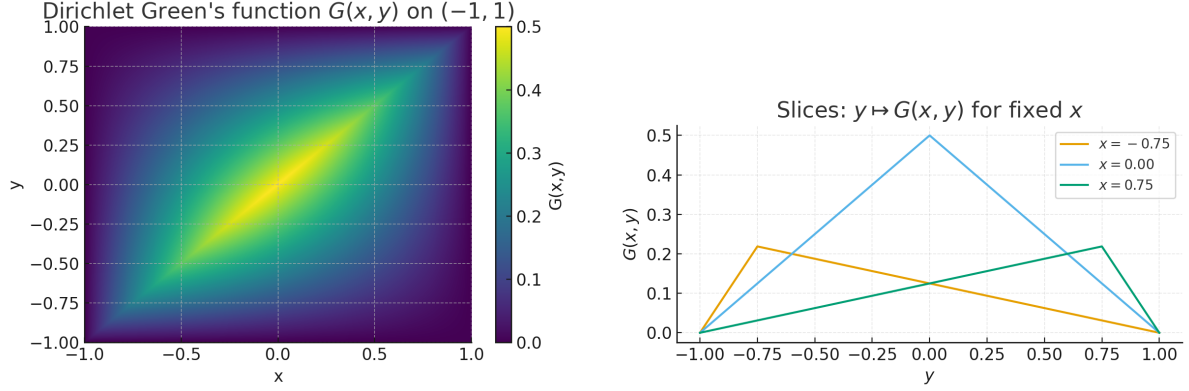
which yields after a straightforward split of the integral at $y = x$:

$$u(x) = \begin{cases} \frac{1+x}{2}, & -1 \leq x < -\frac{1}{2}, \\ -\frac{x^2}{2} + \frac{3}{8}, & |x| \leq \frac{1}{2}, \\ \frac{1-x}{2}, & \frac{1}{2} < x \leq 1. \end{cases}$$

Hence $u \in H_0^1(-1, 1)$, u' is continuous and piecewise linear, and in the distributional sense

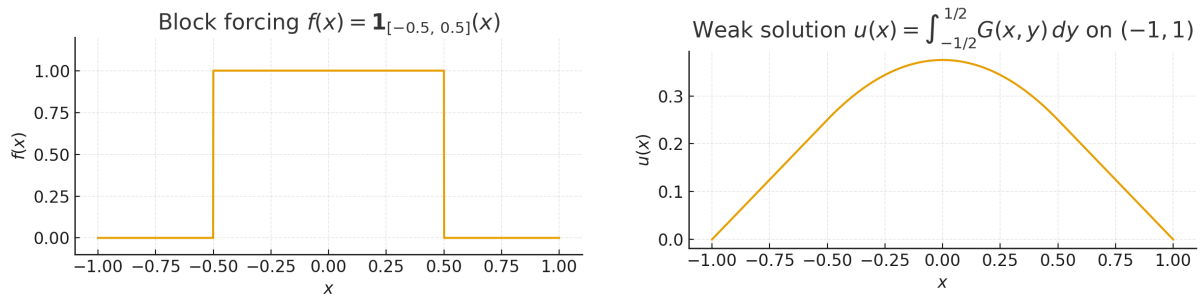
$$u'' = -\mathbf{1}_{[-1/2, 1/2]}.$$

Figures. *Interpretation.* The heatmap and slices display the triangular structure of G . Convolution with the block forcing integrates those triangles over $y \in [-\frac{1}{2}, \frac{1}{2}]$, producing a piecewise quadratic u that is linear outside the support of f and concave on it.



(a) Dirichlet Green's function $G(x, y)$ on $(-1, 1)$. (b) Slices $y \mapsto G(x, y)$ for fixed $x \in \{-0.75, 0, 0.75\}$.

Figure 7: Green's function visualization. The left panel shows the triangular structure in (x, y) ; the right panel shows linear pieces meeting at $y = x$, consistent with $-\partial_{xx}G(\cdot, y) = \delta_y$.



(a) Block forcing $f(x) = \mathbf{1}_{[-0.5, 0.5]}(x)$. (b) Weak solution $u(x) = \int_{-1/2}^{1/2} G(x, y) dy$.

Figure 8: Forcing vs. solution (side-by-side). Convolution of G with the block forcing yields a piecewise quadratic u : linear outside the support of f and concave on it, with $u(\pm 1) = 0$.

Exercise 10 — Weak-* limits of $\sin(nx)$ and $\sin^2(nx)$

Recall that $L^\infty(\mathbb{R}) = (L^1(\mathbb{R}))'$. Consider the sequence $(f_n)_{n \geq 1} \subset L^\infty(\mathbb{R})$ given by

$$f_n(x) = \sin(nx).$$

Show that $f_n \xrightarrow{*} 0$ in $L^\infty(\mathbb{R})$. Show further that $f_n^2 \xrightarrow{*} g$ for some $g \in L^\infty(\mathbb{R})$, and determine g .

Set-up. Identify $L^\infty(\mathbb{R})$ with $(L^1(\mathbb{R}))'$ via the pairing $\langle g, \varphi \rangle = \int_{\mathbb{R}} g(x) \varphi(x) dx$ for $\varphi \in L^1$. Let $f_n(x) = \sin(nx)$.

(a) $f_n \xrightarrow{*} 0$. For any $\varphi \in L^1(\mathbb{R})$,

$$\int_{\mathbb{R}} \sin(nx) \varphi(x) dx \xrightarrow{n \rightarrow \infty} 0$$

by the Riemann–Lebesgue lemma. Hence $f_n \xrightarrow{*} 0$ in L^∞ .

(b) $f_n^2 \xrightarrow{*} \frac{1}{2}$. Using $\sin^2(nx) = \frac{1}{2}(1 - \cos(2nx))$,

$$\int \sin^2(nx) \varphi = \frac{1}{2} \int \varphi - \frac{1}{2} \int \varphi(x) \cos(2nx) dx \longrightarrow \frac{1}{2} \int \varphi(x) dx.$$

Therefore $f_n^2 \xrightarrow{*} g$ with $g(x) \equiv \frac{1}{2}$ in $L^\infty(\mathbb{R})$.

Exercise 11 — A Heisenberg-type inequality and uncertainty

Suppose $f \in \mathcal{S}(\mathbb{R}^n)$. By observing that

$$\|f\|_{L^2}^2 = \int_{\mathbb{R}^n} \frac{1}{n} (\operatorname{div} x) |f(x)|^2 dx,$$

or otherwise, show that

$$(2\pi)^{\frac{n}{2}} \|f\|_{L^2}^2 \leq \frac{2}{n} \left\| |x| f(x) \right\|_{L^2} \left\| |\xi| \hat{f}(\xi) \right\|_{L^2},$$

with equality if and only if $f(x) = a e^{-\lambda|x|^2}$ for some $a \in \mathbb{C}$, $\lambda > 0$. Deduce that for any $x_0, \xi_0 \in \mathbb{R}^n$,

$$(2\pi)^{\frac{n}{2}} \|f\|_{L^2}^2 \leq \frac{2}{n} \left\| |x - x_0| f(x) \right\|_{L^2} \left\| |\xi - \xi_0| \hat{f}(\xi) \right\|_{L^2}.$$

Explain how this shows that a function $f \in L^2(\mathbb{R}^n)$ cannot be sharply localised in both physical and Fourier space simultaneously (the *uncertainty principle*).

Fourier convention. Let

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} f(x) dx, \quad \|f\|_{L^2} = (2\pi)^{-n/2} \|\widehat{f}\|_{L^2}.$$

Step 1: Integration by parts. Since $\operatorname{div} x = n$ and $f \in \mathcal{S}(\mathbb{R}^n)$,

$$\|f\|_2^2 = \frac{1}{n} \int_{\mathbb{R}^n} (\operatorname{div} x) |f|^2 = \frac{2}{n} \Re \int_{\mathbb{R}^n} x \cdot f \overline{\nabla f} dx.$$

Step 2: Cauchy–Schwarz and Plancherel. By Cauchy–Schwarz,

$$\|f\|_2^2 \leq \frac{2}{n} \| |x| f \|_2 \|\nabla f\|_2.$$

Plancherel with the above normalization gives

$$\|\nabla f\|_2 = (2\pi)^{-n/2} \| |\xi| \widehat{f} \|_2,$$

hence

$$(2\pi)^{\frac{n}{2}} \|f\|_2^2 \leq \frac{2}{n} \| |x| f \|_2 \| |\xi| \widehat{f} \|_2. \quad (*)$$

Equality case. Equality in Cauchy–Schwarz requires $\nabla f(x) = -\lambda x f(x)$ for some $\lambda > 0$ (up to a constant phase), whose solutions are precisely

$$f(x) = a e^{-\lambda |x|^2}, \quad a \in \mathbb{C}, \lambda > 0.$$

Shifted form. Applying $(*)$ to $g(x) = e^{-i\xi_0 \cdot x} f(x - x_0)$ yields

$$(2\pi)^{\frac{n}{2}} \|f\|_2^2 \leq \frac{2}{n} \| |x - x_0| f \|_2 \| |\xi - \xi_0| \widehat{f} \|_2, \quad x_0, \xi_0 \in \mathbb{R}^n.$$

Uncertainty principle (interpretation). The product of the L^2 -spreads in physical and Fourier space controls $\|f\|_2^2$. One cannot make both $\| |x - x_0| f \|_2$ and $\| |\xi - \xi_0| \widehat{f} \|_2$ arbitrarily small unless $f \equiv 0$. Gaussians are the unique (up to symmetries) extremizers.

Exercise 12 — Sign, Heaviside, and principal values

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the sign function

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0, \end{cases} \quad \text{and} \quad f_R(x) = f(x) \mathbf{1}_{[-R, R]}(x).$$

(a) Sketch $f_R(x)$, and show that $T_{f_R} \rightarrow T_f$ in $\mathcal{S}'(\mathbb{R})$ as $R \rightarrow \infty$.

(b) Compute $\widehat{f_R}(\xi)$, and show that for $\phi \in \mathcal{S}(\mathbb{R})$,

$$T_{f_R}[\phi] = -2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x} dx + 2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x} \cos(Rx) dx.$$

Deduce that $\widehat{T_f} = -2i \text{P.V.}\left(\frac{1}{\xi}\right)$, where the principal value distribution

$$\text{P.V.}\left(\frac{1}{x}\right)[\phi] = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R} \setminus (-\varepsilon, \varepsilon)} \frac{\phi(x)}{x} dx, \quad \phi \in \mathcal{S}(\mathbb{R}),$$

is understood.

(c) Write down $\widehat{T_H}$, where H is the Heaviside function

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

By considering $e^{-\varepsilon x} H(x)$, or otherwise, find an expression for the distribution u defined by

$$u[\phi] := \lim_{\varepsilon \downarrow 0^+} \int_{\mathbb{R}} \frac{\phi(x)}{x + i\varepsilon} dx, \quad \phi \in \mathcal{S}(\mathbb{R}).$$

Solution Let $f(x) = \text{sgn } x$ and $f_R(x) = f(x) \mathbf{1}_{[-R, R]}(x)$.

(a) $T_{f_R} \rightarrow T_f$ in $\mathcal{S}'(\mathbb{R})$. For any $\phi \in \mathcal{S}(\mathbb{R})$,

$$\langle T_{f_R} - T_f, \phi \rangle = \int_{|x| > R} f(x) \phi(x) dx \longrightarrow 0$$

by rapid decay of ϕ .

(b) $\widehat{T_f} = -2i \text{ p.v.}(1/\xi)$. For $\xi \neq 0$,

$$\widehat{f_R}(\xi) = \int_{-R}^R e^{-ix\xi} \text{sgn } x \, dx = -\frac{2i}{\xi} (1 - \cos(R\xi)).$$

Hence, for $\phi \in \mathcal{S}$,

$$\begin{aligned} \langle \widehat{T_{f_R}}, \phi \rangle &= \langle T_{f_R}, \widehat{\phi} \rangle = -2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x} \, dx + 2i \int_0^\infty \frac{\phi(x) - \phi(-x)}{x} \cos(Rx) \, dx \\ &\xrightarrow{R \rightarrow \infty} -2i \text{ p.v.} \int_{\mathbb{R}} \frac{\phi(x)}{x} \, dx. \end{aligned}$$

Thus $\widehat{T_f} = -2i \text{ p.v.} \frac{1}{\xi}$.

(c) $\widehat{T_H}$ and the distribution $u = \lim_{\varepsilon \downarrow 0} \frac{1}{x+i\varepsilon}$. Since $H = (\text{sgn} + 1)/2$,

$$\widehat{T_H} = \frac{1}{2} \widehat{T_{\text{sgn}}} + \frac{1}{2} \widehat{1} = -i \text{ p.v.} \frac{1}{\xi} + \pi \delta.$$

Moreover,

$$\frac{1}{x+i0} := \lim_{\varepsilon \downarrow 0} \frac{1}{x+i\varepsilon} = \text{p.v.} \frac{1}{x} - i\pi \delta_0,$$

i.e.

$$u[\phi] = \lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}} \frac{\phi(x)}{x+i\varepsilon} \, dx = \text{p.v.} \int_{\mathbb{R}} \frac{\phi(x)}{x} \, dx - i\pi \phi(0).$$

Exercise 13 — Slices of a distribution and smooth approximation

Suppose $\phi \in C_c^\infty(\mathbb{R}^n \times \mathbb{R}^m)$. For each $y \in \mathbb{R}^m$ let $\phi_y : \mathbb{R}^n \rightarrow \mathbb{C}$ be given by $\phi_y(x) = \phi(x, y)$. Let $u \in \mathcal{D}'(\mathbb{R}^n)$.

(a) Show that the map $\psi : \mathbb{R}^m \rightarrow \mathbb{C}$, $\psi(y) := u[\phi_y]$, is smooth and obtain a formula for $D_y^\alpha \psi(y)$. Deduce that

$$\int_{\mathbb{R}^m} \psi(y) \, dy = u[\Psi], \quad \text{where } \Psi(x) := \int_{\mathbb{R}^m} \phi(x, y) \, dy.$$

(b) Show that there exists a sequence of smooth, compactly supported functions $f_n \in C_c^\infty(\mathbb{R}^n)$ such that $T_{f_n} \rightarrow u$ in the weak-* topology of $\mathcal{D}'(\mathbb{R}^n)$.

Solution Let $\phi \in C_c^\infty(\mathbb{R}^n \times \mathbb{R}^m)$ and, for $y \in \mathbb{R}^m$, define $\phi_y(x) = \phi(x, y)$. Let $u \in \mathcal{D}'(\mathbb{R}^n)$.

(a) **Smoothness of $\psi(y) = u[\phi_y]$ and formula for derivatives.** The map $y \mapsto \phi_y$ is C^∞ into $\mathcal{D}(\mathbb{R}^n)$, hence

$$D_y^\beta \psi(y) = u[D_y^\beta \phi_y] = u[(\partial_y^\beta \phi)(\cdot, y)].$$

By Fubini and compact support,

$$\int_{\mathbb{R}^m} \psi(y) dy = u\left[\int_{\mathbb{R}^m} \phi(\cdot, y) dy\right] =: u[\Psi], \quad \Psi(x) = \int_{\mathbb{R}^m} \phi(x, y) dy.$$

(b) **Approximation by C_c^∞ functions.** Choose $\rho \in C_c^\infty(\mathbb{R}^n)$ with $\int \rho = 1$ and set $\rho_\varepsilon(x) = \varepsilon^{-n} \rho(x/\varepsilon)$. Let $u_\varepsilon = u * \rho_\varepsilon \in C^\infty$ and pick $\chi_R \in C_c^\infty$ with $\chi_R \equiv 1$ on $B(0, R)$. Define $f_{R,\varepsilon} = \chi_R u_\varepsilon \in C_c^\infty$. Then for any $\varphi \in \mathcal{D}$ and all R large (so that $\chi_R \equiv 1$ on $\text{supp } \varphi$),

$$\int f_{R,\varepsilon} \varphi = \int u_\varepsilon \varphi \xrightarrow{\varepsilon \rightarrow 0} u[\varphi].$$

A diagonal choice f_n yields $T_{f_n} \rightarrow u$ in the weak-* topology of $\mathcal{D}'(\mathbb{R}^n)$.

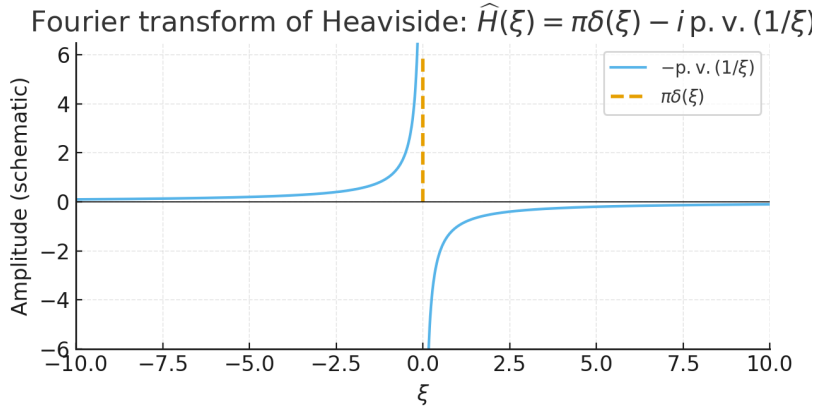


Figure 9: Fourier transform of the Heaviside function. The real part is $\pi\delta(\xi)$, shown as the vertical spike, and the imaginary part $-i \text{p.v.}(1/\xi)$ appears as an odd $1/\xi$ tail. This decomposition illustrates the distributional nature of $\widehat{H}$.

Exercise 1

Suppose $f \in L_{\text{loc}}^p(\mathbb{R}^n)$ is $\mathbb{Z}^n$ -periodic and let $q = \{x \in \mathbb{R}^n : |x_j| < \frac{1}{2}, j = 1, \dots, n\}$. Show that for any $\varepsilon > 0$ there exists a smooth periodic function f_ε such that $\|f_\varepsilon - f\|_{L^p(q)} < \varepsilon$.

Solution. Let $\rho \in C_c^\infty(\mathbb{R}^n)$ be a standard mollifier, $\rho_\delta(x) = \delta^{-n} \rho(x/\delta)$. Define the periodic convolution on the torus $\mathbb{T}^n$ by

$$f_\delta(x) := (f * \mathbb{T}^n \rho_\delta)(x) = \int_q f(x - y) \rho_\delta(y) dy.$$

Then $f_\delta \in C^\infty(\mathbb{R}^n)$ is 1-periodic and $f_\delta \rightarrow f$ in $L^p(\mathbb{T}^n)$ as $\delta \downarrow 0$. Equivalently, $\|f_\delta - f\|_{L^p(q)} \rightarrow 0$. Given $\varepsilon > 0$, choose δ small and set $f_\varepsilon := f_\delta$.

Exercise 2

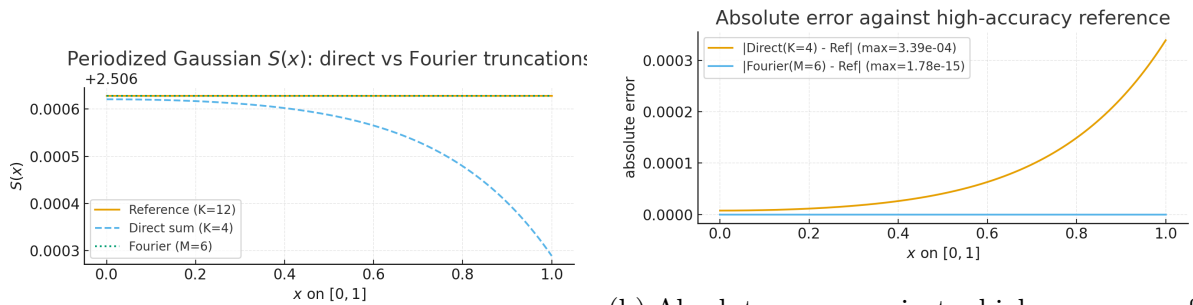
Show that the series $S(x) = \sum_{k \in \mathbb{Z}^n} e^{-|x+k|^2/2}$ converges for every $x \in (0, 1]^n$ and compute the Fourier coefficients of S (with the convention $e^{2\pi i m \cdot x}$, $m \in \mathbb{Z}^n$).

Solution. The summand $e^{-|x+k|^2/2}$ decays super-exponentially in $|k|$, so the series converges absolutely and defines a smooth 1-periodic function on $\mathbb{T}^n$. For $m \in \mathbb{Z}^n$,

$$\hat{S}(m) = \int_q S(x) e^{-2\pi i m \cdot x} dx = \sum_{k \in \mathbb{Z}^n} \int_q e^{-|x+k|^2/2} e^{-2\pi i m \cdot x} dx = \int_{\mathbb{R}^n} e^{-|y|^2/2} e^{-2\pi i m \cdot y} dy.$$

Since $\int_{\mathbb{R}^n} e^{-|y|^2/2} e^{-2\pi i m \cdot y} dy = (2\pi)^{n/2} e^{-2\pi^2 |m|^2}$, the Fourier series is

$$S(x) = \sum_{m \in \mathbb{Z}^n} (2\pi)^{n/2} e^{-2\pi^2 |m|^2} e^{2\pi i m \cdot x}.$$



(a) Direct sum vs. Fourier truncation for $S(x)$. (b) Absolute errors against a high-accuracy reference.

Figure 10: Numerical check for the periodized Gaussian $S(x) = \sum_{k \in \mathbb{Z}} e^{-(x+k)^2/2}$: the Fourier coefficients $(2\pi)^{1/2} e^{-2\pi^2 m^2}$ yield rapid spectral convergence.

Exercise 3

Suppose $\Omega \subset \mathbb{R}^n$ is open and bounded, $f \in C_c^\infty(\Omega)$, and $0 < \varepsilon < 1$.

(a) Show that $\int_{\Omega} (|f|^2 + \varepsilon)^{p/2} dx \rightarrow \|f\|_{L^p(\Omega)}^p$ as $\varepsilon \downarrow 0$.

Solution. Pointwise $(|f|^2 + \varepsilon)^{p/2} \rightarrow |f|^p$ and $(|f|^2 + \varepsilon)^{p/2} \leq (|f|^2 + 1)^{p/2} \in L^1(\Omega)$. Dominated convergence yields the claim.

(b) Show that there exists $C = C(\Omega, p)$ such that $\|f\|_{L^p(\Omega)} \leq C \|Df\|_{L^p(\Omega)}$.

Solution. Extend f by 0 outside Ω and set $F(x) = \frac{1}{n} x (|f|^2 + \varepsilon)^{p/2}$. Then $\int_{\mathbb{R}^n} \operatorname{div} F \, dx = 0$, hence

$$\int_{\mathbb{R}^n} (|f|^2 + \varepsilon)^{p/2} dx = -\frac{1}{n} \int_{\mathbb{R}^n} x \cdot \nabla (|f|^2 + \varepsilon)^{p/2} dx.$$

Using $|\nabla(|f|^2)| \leq 2|f||\nabla f|$,

$$\int (|f|^2 + \varepsilon)^{p/2} \leq \frac{p}{n} \int |x| (|f|^2 + \varepsilon)^{\frac{p}{2}-1} |f| |\nabla f| \leq C_{\Omega} \int (|f|^2 + \varepsilon)^{\frac{p-1}{2}} |\nabla f|.$$

Hölder with exponents p and $p' = \frac{p}{p-1}$ gives

$$\int (|f|^2 + \varepsilon)^{\frac{p}{2}} \leq C_{\Omega} \left(\int (|f|^2 + \varepsilon)^{\frac{p}{2}} \right)^{\frac{p-1}{p}} \|\nabla f\|_{L^p},$$

where $C_{\Omega} = \frac{p}{n} \sup_{x \in \Omega} |x|$. Hence

$$\left(\int (|f|^2 + \varepsilon)^{\frac{p}{2}} \right)^{1/p} \leq C_{\Omega} \|\nabla f\|_{L^p}.$$

Letting $\varepsilon \downarrow 0$ and using part (a) yields

$$\|f\|_{L^p(\Omega)} \leq C \|\nabla f\|_{L^p(\Omega)},$$

with C depending only on Ω and p .

Exercise 4

Let $s \in \mathbb{R}$.

(a) Show that $\mathcal{S}(\mathbb{R}^n)$ is a dense subset of $H^s(\mathbb{R}^n)$.

- (b) Find a condition on s such that the Dirac mass δ_x belongs to $H^s(\mathbb{R}^n)$.
(c) Show that $H^t(\mathbb{R}^n)$ is continuously embedded in $H^s(\mathbb{R}^n)$ whenever $s < t$.
(d) Show that the derivative operator D^α is a bounded linear map

$$D^\alpha : H^{s+k}(\mathbb{R}^n) \rightarrow H^s(\mathbb{R}^n), \quad k = |\alpha|.$$

- (e) (*) Show that the pairing

$$\langle f, g \rangle = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(\xi) \widehat{g}(\xi) d\xi, \quad f \in H^{-s}(\mathbb{R}^n), \quad g \in H^s(\mathbb{R}^n),$$

is well defined, and that the map $g \mapsto \langle f, g \rangle$ is a bounded linear operator on $H^s(\mathbb{R}^n)$. Deduce that $(H^s(\mathbb{R}^n))' \simeq H^{-s}(\mathbb{R}^n)$, and that $\mathcal{S}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$ for all s .

Solution. Let $s \in \mathbb{R}$ and define

$$\|f\|_{H^s(\mathbb{R}^n)}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi.$$

(a) $\mathcal{S}$ is dense in $H^s(\mathbb{R}^n)$. For $f \in H^s$, set $\widehat{f_R} = \chi_{B(0,R)} \widehat{f}$. Then $f_R \rightarrow f$ in H^s as $R \rightarrow \infty$. Let $\rho \in \mathcal{S}$ be a mollifier and put $f_{R,\varepsilon} = f_R * \rho_\varepsilon \in \mathcal{S}$. One has $\|f_{R,\varepsilon} - f_R\|_{H^s} \rightarrow 0$ as $\varepsilon \downarrow 0$. A diagonal choice gives a sequence in $\mathcal{S}$ converging to f in H^s .

(b) Condition for $\delta_x \in H^s(\mathbb{R}^n)$. Since $\widehat{\delta_x}(\xi) = e^{-ix \cdot \xi}$,

$$\|\delta_x\|_{H^s}^2 \sim \int_{\mathbb{R}^n} (1 + |\xi|^2)^s d\xi < \infty \iff s < -\frac{n}{2}.$$

(c) Continuous embedding. If $s < t$ then $(1 + |\xi|^2)^s \leq (1 + |\xi|^2)^t$, hence $H^t(\mathbb{R}^n) \hookrightarrow H^s(\mathbb{R}^n)$ and $\|f\|_{H^s} \leq \|f\|_{H^t}$.

(d) Derivatives act boundedly. For $\alpha \in \mathbb{N}^n$, $k = |\alpha|$,

$$\|D^\alpha f\|_{H^s}^2 = \int (1 + |\xi|^2)^s |\xi|^{2k} |\widehat{f}|^2 \leq \int (1 + |\xi|^2)^{s+k} |\widehat{f}|^2 = \|f\|_{H^{s+k}}^2.$$

(e) Duality and identifications. For $f \in H^{-s}$ and $g \in H^s$ define

$$\langle f, g \rangle = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(\xi) \overline{\widehat{g}(\xi)} d\xi.$$

By Cauchy-Schwarz with the weights $(1 + |\xi|^2)^{-s/2}$ and $(1 + |\xi|^2)^{s/2}$ the pairing is well-defined and bounded in g . Thus $(H^s)'$ is identified with H^{-s} . In particular, $\mathcal{S}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$ for all s .

Exercise 5

For two probability measures μ, ν on $\mathbb{R}^n$, define their convolution by

$$(\mu * \nu)(\phi) = \iint_{\mathbb{R}^n} \phi(x+y) d\mu(x) d\nu(y), \quad \phi \in \mathcal{S}(\mathbb{R}^n).$$

- (i) Show that $\mu * \nu$ defines an element $T_{\mu*\nu} \in \mathcal{S}'(\mathbb{R}^n)$ and that

$$\widehat{T_{\mu*\nu}} = \widehat{T_\mu} \widehat{T_\nu}.$$

- (ii) Let $n = 1$ and let μ be a probability measure on $\mathbb{R}$ with $\int x d\mu(x) = 0$, $\int x^2 d\mu(x) = 1$. Let μ^{*k} denote the k -fold convolution of μ with itself, and define a scaled probability measure $\lambda_k(A) = \mu^{*k}(\sqrt{k} A)$ for $A \subset \mathbb{R}$ Borel. Show that the corresponding distributions T_{λ_k} converge in $\mathcal{S}'(\mathbb{R})$ as $k \rightarrow \infty$ and identify the limit.
- (iii) (*) Show that T_{λ_k} defines a sequence in $H^{-s}(\mathbb{R})$ whenever $s > \frac{1}{2}$ and that it converges in $H^{-s}(\mathbb{R})$ to the same limit. What happens if $s \leq \frac{1}{2}$?

Solution. For probability measures μ, ν on $\mathbb{R}^n$ define their convolution by

$$(\mu * \nu)(\phi) = \iint_{\mathbb{R}^n} \phi(x+y) d\mu(x) d\nu(y), \quad \phi \in \mathcal{S}(\mathbb{R}^n).$$

- (i) $\widehat{T_{\mu*\nu}} = \widehat{T_\mu} \widehat{T_\nu}$. For $\xi \in \mathbb{R}^n$,

$$\widehat{T_{\mu*\nu}}(\xi) = \int e^{-i\xi \cdot z} d(\mu * \nu)(z) = \iint e^{-i\xi \cdot (x+y)} d\mu(x) d\nu(y) = \widehat{T_\mu}(\xi) \widehat{T_\nu}(\xi).$$

- (ii) **Central limit rescaling on $\mathbb{R}$.** Assume $\int x d\mu = 0$, $\int x^2 d\mu = 1$, set $\lambda_k(A) = \mu^{*k}(\sqrt{k} A)$. Then

$$\widehat{T_{\lambda_k}}(\xi) = \left(\widehat{T_\mu}(\xi/\sqrt{k}) \right)^k \xrightarrow{k \rightarrow \infty} e^{-\xi^2/2}.$$

Hence $T_{\lambda_k} \rightarrow T_\gamma$ in $\mathcal{S}'(\mathbb{R})$, where T_γ is the standard Gaussian distribution.

- (iii) **Convergence in $H^{-s}(\mathbb{R})$.** For $s > 1/2$,

$$\|T_{\lambda_k}\|_{H^{-s}}^2 \sim \int_{\mathbb{R}} (1 + \xi^2)^{-s} |\widehat{T_{\lambda_k}}(\xi)|^2 d\xi \leq \int_{\mathbb{R}} (1 + \xi^2)^{-s} d\xi < \infty,$$

and dominated convergence implies $T_{\lambda_k} \rightarrow T_\gamma$ in H^{-s} . If $s \leq 1/2$, the weight $(1 + \xi^2)^{-s}$ is not integrable at infinity, so the sequence need not be bounded in H^{-s} and convergence in H^{-s} fails in general.

Exercise 6

Let $s = \frac{n}{2} + \gamma$ with $0 < \gamma < 1$. Show that there exists $C_{n,\gamma} > 0$ such that for all $x, y \in \mathbb{R}^n$,

$$\int_{\mathbb{R}^n} \frac{|e^{ix \cdot \xi} - e^{iy \cdot \xi}|^2}{|\xi|^{2s}} d\xi \leq C_{n,\gamma} |x - y|^{2\gamma}.$$

Solution. With $h = x - y$,

$$|e^{ix \cdot \xi} - e^{iy \cdot \xi}|^2 = 2(1 - \cos(h \cdot \xi)), \quad 2s = n + 2\gamma.$$

Hence

$$\int_{\mathbb{R}^n} \frac{|e^{ix \cdot \xi} - e^{iy \cdot \xi}|^2}{|\xi|^{2s}} d\xi = 2 \int_{\mathbb{R}^n} \frac{1 - \cos(h \cdot \xi)}{|\xi|^{n+2\gamma}} d\xi.$$

Use $1 - \cos t \leq \min\{t^2, 2\}$, rotate so that $h = |h|e_1$, and split at $|\xi| = |h|^{-1}$. For $|\xi| \leq |h|^{-1}$,

$$1 - \cos(|h|\xi_1) \lesssim |h|^2 \xi_1^2 \quad \Rightarrow \quad \int_{|\xi| \leq |h|^{-1}} \frac{1 - \cos(|h|\xi_1)}{|\xi|^{n+2\gamma}} \lesssim |h|^2 \int_{|\xi| \leq |h|^{-1}} \frac{\xi_1^2}{|\xi|^{n+2\gamma}} \sim |h|^{2\gamma}.$$

For $|\xi| > |h|^{-1}$,

$$1 - \cos(\cdot) \leq 2 \quad \Rightarrow \quad \int_{|\xi| > |h|^{-1}} \frac{1}{|\xi|^{n+2\gamma}} \sim |h|^{2\gamma}.$$

Combining yields the stated bound with a constant $C_{n,\gamma}$.

Sobolev–Bessel spaces $H^s(\mathbb{R}^n)$

Fourier/Bessel definition (all $s \in \mathbb{R}$). For $f \in \mathcal{S}'(\mathbb{R}^n)$ with Fourier transform $\widehat{f}$,

$$\|f\|_{H^s(\mathbb{R}^n)}^2 := \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi < \infty.$$

Equivalently, $f \in H^s$ iff $(I - \Delta)^{s/2} f \in L^2$, and $\|f\|_{H^s} = \|(I - \Delta)^{s/2} f\|_{L^2}$. For $s = k \in \mathbb{N}$ one has $H^k(\mathbb{R}^n) = W^{k,2}(\mathbb{R}^n)$.

Fractional characterization ($0 < s < 1$). There holds the norm equivalence

$$\|f\|_{H^s}^2 \asymp \|f\|_{L^2}^2 + \iint_{\mathbb{R}^n \times \mathbb{R}^n} \frac{|f(x) - f(y)|^2}{|x - y|^{n+2s}} dx dy,$$

where the right-hand side defines the Gagliardo seminorm $[f]_{H^s}^2 := \iint \frac{|f(x) - f(y)|^2}{|x - y|^{n+2s}} dx dy$.

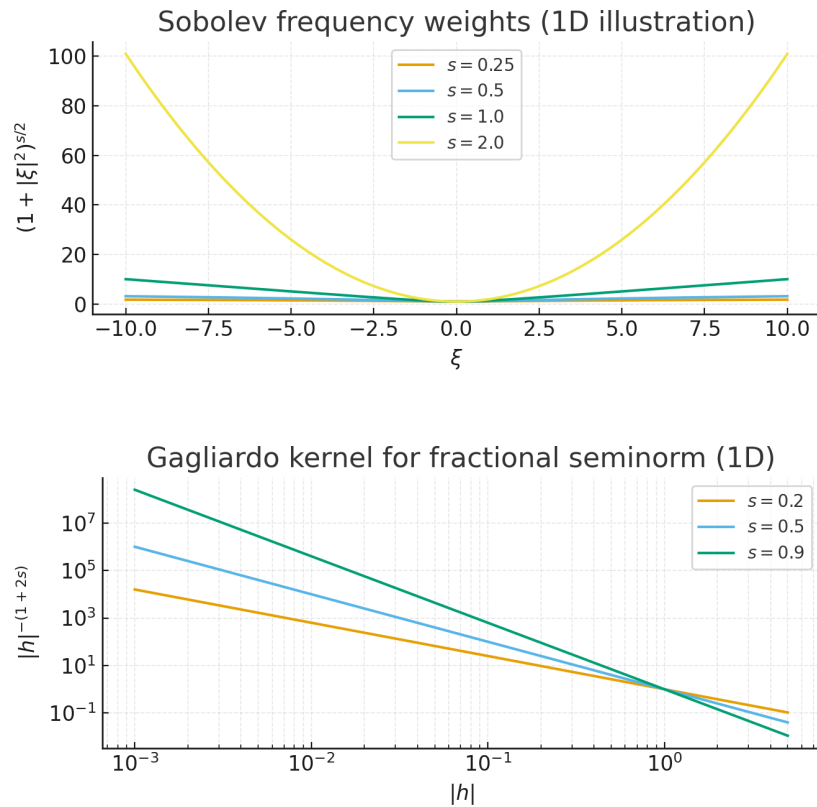
Basic properties. $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$ for all s . If $t > s$ then $H^t(\mathbb{R}^n) \hookrightarrow H^s(\mathbb{R}^n)$ continuously. For $\alpha \in \mathbb{N}^n$, $D^\alpha : H^{s+|\alpha|} \rightarrow H^s$ is bounded. The dual space $(H^s(\mathbb{R}^n))'$ identifies with $H^{-s}(\mathbb{R}^n)$ via

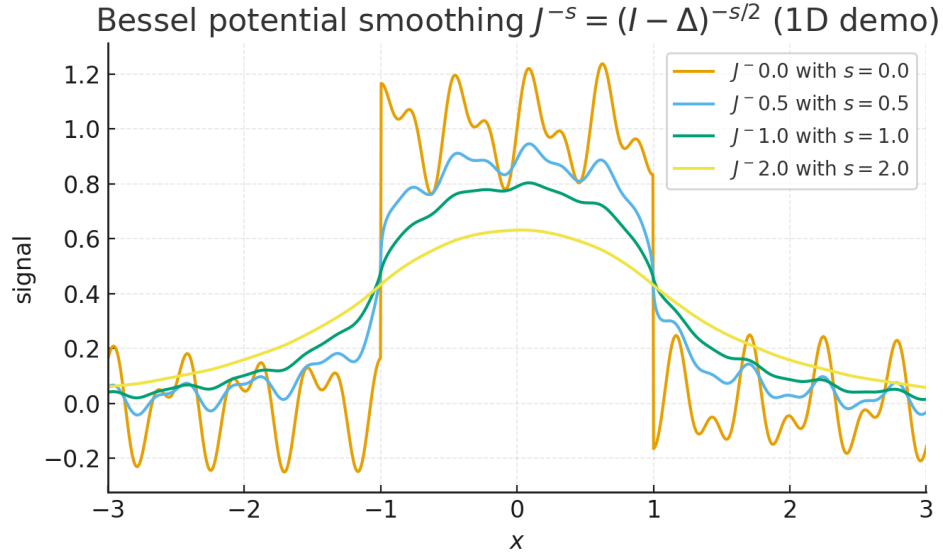
$$\langle f, g \rangle = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{f}(\xi) \overline{\widehat{g}(\xi)} d\xi.$$

Sobolev embeddings. If $s > n/2$ then $H^s(\mathbb{R}^n) \hookrightarrow C^{0,\alpha}(\mathbb{R}^n)$ for any $\alpha < s - n/2$. Moreover, $H^s(\mathbb{R}^n)$ is a Banach algebra for $s > n/2$: $\|fg\|_{H^s} \lesssim \|f\|_{H^s} \|g\|_{H^s}$.

Rellich compactness and traces. If $\Omega \Subset \mathbb{R}^n$ and $t < s$, then $H^s(\Omega) \hookrightarrow H^t(\Omega)$ compactly. If $s > 1/2$ and $\Omega = \mathbb{R}_+^n$, the trace operator is bounded, $H^s(\mathbb{R}_+^n) \rightarrow H^{s-1/2}(\partial\mathbb{R}_+^n)$.

Figures (1D illustrations).





Interpretation. Top: frequency weights $(1 + |\xi|^2)^{s/2}$ increase with s , penalising high frequencies. Middle: the Gagliardo kernel $|h|^{-(n+2s)}$ becomes more singular as $s \uparrow 1$, enforcing stronger control of increments $f(x+h) - f(x)$. Bottom: the Bessel potential $J^{-s} = (I - \Delta)^{-s/2}$ acts as a low-pass filter and smooths a rough signal increasingly with s .

Example A (Gaussian is in all H^s). Let $g(x) = e^{-|x|^2/2}$ on $\mathbb{R}^n$. Since $\widehat{g}(\xi) = (2\pi)^{n/2}e^{-|\xi|^2/2}$,

$$\|g\|_{H^s(\mathbb{R}^n)}^2 = (2\pi)^n \int_{\mathbb{R}^n} (1 + |\xi|^2)^s e^{-|\xi|^2} d\xi < \infty$$

for every $s \in \mathbb{R}$, hence $g \in H^s(\mathbb{R}^n)$ for all s .

Example B (Indicator of an interval; threshold $s < \frac{1}{2}$ in 1D). Let $f = \mathbf{1}_{(0,1)}$ on $\mathbb{R}$. Then

$$\widehat{f}(\xi) = e^{-i\xi/2} \frac{2 \sin(\xi/2)}{\xi}, \quad |\widehat{f}(\xi)|^2 \sim \frac{4}{\xi^2} \quad (|\xi| \rightarrow \infty).$$

Therefore

$$\|f\|_{H^s(\mathbb{R})}^2 \sim \int_{\mathbb{R}} (1 + \xi^2)^s \frac{1}{\xi^2} d\xi < \infty \iff s < \frac{1}{2}.$$

Equivalently, for $0 < s < 1$,

$$[f]_{H^s(\mathbb{R})}^2 = \iint_{\mathbb{R}^2} \frac{|f(x) - f(y)|^2}{|x - y|^{1+2s}} dx dy < \infty \iff s < \frac{1}{2},$$

since the only singular contribution comes from pairs (x, y) straddling the endpoints.

Example C (Power cusp on a bounded interval). Let $f(x) = |x|^\beta \chi_{[-1,1]}(x)$ with $\beta > -1/2$ (so $f \in L^2$). Then

$$f \in H^s((-1, 1)) \iff s < \beta + \frac{1}{2}.$$

Indeed, near 0,

$$\iint_{(0,1)^2} \frac{|x^\beta - y^\beta|^2}{|x - y|^{1+2s}} dx dy < \infty \iff s < \beta + \frac{1}{2},$$

and outside a neighbourhood of 0 the integrand is smooth. A Fourier-tail argument gives the same threshold.

Remark (Periodic case). On $\mathbb{T}$, for $u_m(x) = \sin(2\pi mx)$ one has $\|u_m\|_{H^s(\mathbb{T})} \asymp (1 + m^2)^{s/2}$, which exhibits directly the frequency weight $(1 + |m|^2)^{s/2}$.

Example A (Gaussian in all H^s). Let $g(x) = e^{-|x|^2/2}$ on $\mathbb{R}^n$. Then $\widehat{g}(\xi) = (2\pi)^{n/2}e^{-|\xi|^2/2}$ and

$$\|g\|_{H^s}^2 = (2\pi)^n \int_{\mathbb{R}^n} (1 + |\xi|^2)^s e^{-|\xi|^2} d\xi < \infty \quad \forall s \in \mathbb{R}.$$

Hence $g \in H^s(\mathbb{R}^n)$ for every s .

Example B (Indicator; threshold $s < \frac{1}{2}$ in 1D). For $f = \mathbf{1}_{(0,1)}$ one has $\hat{f}(\xi) = e^{-i\xi/2} \frac{2\sin(\xi/2)}{\xi}$, so $|\hat{f}(\xi)|^2 \sim 2/\xi^2$ as $|\xi| \rightarrow \infty$. Therefore

$$\|f\|_{H^s(\mathbb{R})}^2 \sim \int_{\mathbb{R}} (1 + \xi^2)^s \frac{1}{\xi^2} d\xi < \infty \iff s < \frac{1}{2}.$$

Equivalently, for $0 < s < 1$, the Gagliardo seminorm $\iint \frac{|f(x)-f(y)|^2}{|x-y|^{1+2s}} dx dy$ converges iff $s < \frac{1}{2}$.

Example C (Smooth bump in all H^s). Let $b(x) = e^{-1/(1-x^2)}$ for $|x| < 1$ and $b = 0$ otherwise. Then $b \in C_c^\infty(\mathbb{R})$ and for every N , $|\hat{b}(\xi)| \leq C_N(1 + |\xi|)^{-N}$. Hence $\|b\|_{H^s} < \infty$ for all $s \in \mathbb{R}$.

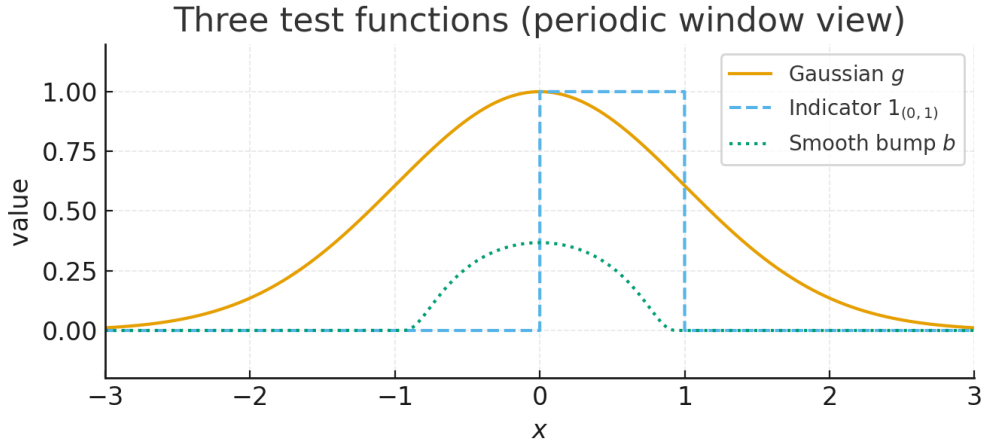


Figure 11: Three test functions in physical space.

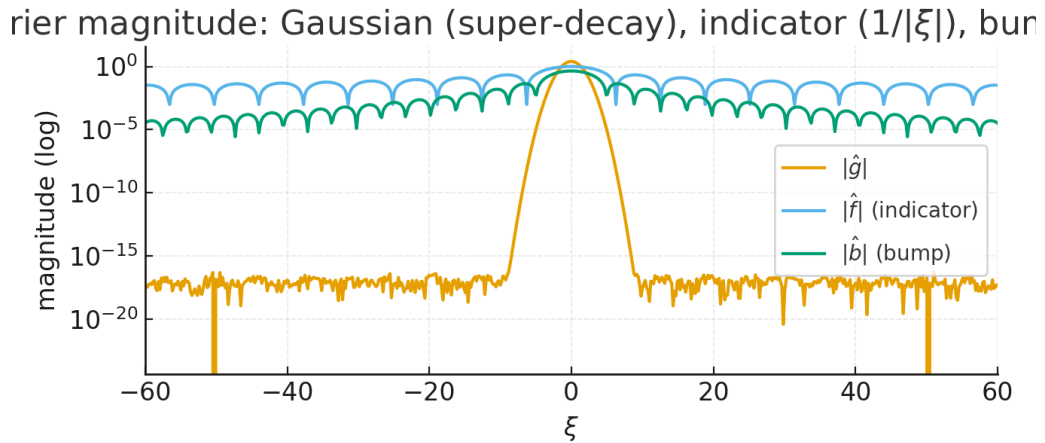


Figure 12: Fourier magnitudes. Gaussian and bump decay super-polynomially; indicator decays like $1/|\xi|$.

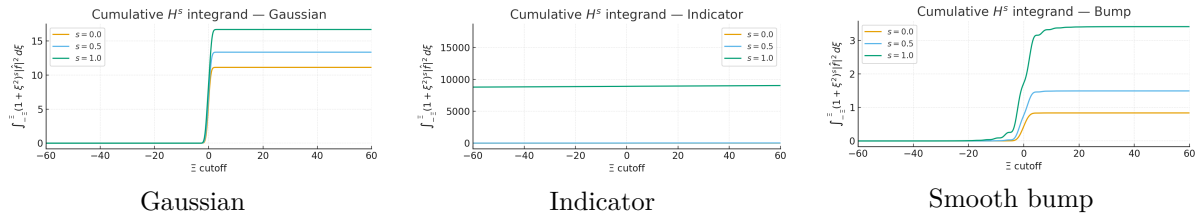


Figure 13: Cumulative Sobolev integrals $\int_{-\Xi}^{\Xi} (1 + \xi^2)^s |\hat{f}(\xi)|^2 d\xi$ for $s \in \{0, 0.5, 1\}$.

Exercise 6(b)

Show that if $s = \frac{n}{2} + k + \gamma$ for some $k \in \mathbb{Z}_{\geq 0}$ and $0 < \gamma < 1$, then

$$H^s(\mathbb{R}^n) \subset C^{k,\gamma}(\mathbb{R}^n).$$

Solution. Let $u \in H^s(\mathbb{R}^n)$ and fix $|\alpha| = k$. The derivative map is bounded, $D^\alpha : H^s \rightarrow H^{s-k} = H^{\frac{n}{2}+\gamma}$. For $0 < \gamma < 1$ the (noninteger) Morrey–Sobolev embedding yields

$$H^{\frac{n}{2}+\gamma}(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n).$$

Hence $D^\alpha u \in C^{0,\gamma}$ for all $|\alpha| = k$, equivalently $u \in C^{k,\gamma}(\mathbb{R}^n)$. Moreover,

$$\|u\|_{C^{k,\gamma}} \lesssim \sum_{|\alpha|=k} \|D^\alpha u\|_{C^{0,\gamma}} \lesssim \sum_{|\alpha|=k} \|D^\alpha u\|_{H^{\frac{n}{2}+\gamma}} \lesssim \|u\|_{H^{\frac{n}{2}+k+\gamma}}.$$

□

Exercise 7

Fix $s \in \mathbb{R}$ and suppose $f \in H^s(\mathbb{R}^n)$.

- (a) Show that there exists a unique $u \in H^{s+4}(\mathbb{R}^n)$ solving

$$\Delta^2 u + u = f \quad \text{in } \mathbb{R}^n.$$

- (b) Prove the a priori bound $\|u\|_{H^{s+4}} \leq C \|f\|_{H^s}$ with C independent of f .
(c) For which s does the equation hold in the sense of classical derivatives (possibly after redefining u, f on a null set)?

Solution. Taking Fourier transforms gives

$$(1 + |\xi|^4) \hat{u}(\xi) = \hat{f}(\xi), \quad \hat{u}(\xi) = m(\xi) \hat{f}(\xi), \quad m(\xi) = \frac{1}{1 + |\xi|^4}.$$

Since $m \in S^{-4}$, one has

$$(1 + |\xi|^2)^{\frac{s+4}{2}} |m(\xi)| \lesssim (1 + |\xi|^2)^{\frac{s}{2}} \Rightarrow \|u\|_{H^{s+4}} \leq C \|f\|_{H^s}.$$

Thus $u \in H^{s+4}$ and the estimate in (b) holds. Uniqueness follows because the symbol $1 + |\xi|^4$ does not vanish: if $f = 0$ then $\hat{u} = 0$.

For (c), by Sobolev–Morrey,

$$H^t(\mathbb{R}^n) \hookrightarrow C^{k,\alpha}(\mathbb{R}^n) \quad \text{whenever } t > \frac{n}{2} + k + \alpha, \quad 0 < \alpha < 1.$$

With $t = s + 4$ and $k = 4$ we get $u \in C^{4,\alpha}$ for some $\alpha > 0$ provided $s + 4 > \frac{n}{2} + 4 + \alpha$, i.e.

$$\boxed{s > \frac{n}{2}}.$$

In that case $\Delta^2 u$ is continuous, $f \in H^s \hookrightarrow C^{0,\alpha}$, and the identity $\Delta^2 u + u = f$ holds pointwise (after modifying on a set of measure zero). For $s \leq \frac{n}{2}$ the solution is generally only a weak (distributional) solution. $\square$

Exercise 6(b)

Suppose

$$s = \frac{n}{2} + k + \gamma, \quad k \in \mathbb{Z}_{\geq 0}, \quad 0 < \gamma < 1.$$

Show that

$$H^s(\mathbb{R}^n) \subset C^{k,\gamma}(\mathbb{R}^n).$$

Solution. Let $u \in H^s(\mathbb{R}^n)$ and fix a multi-index α with $|\alpha| = k$. The derivative map

$$D^\alpha : H^s(\mathbb{R}^n) \longrightarrow H^{s-k}(\mathbb{R}^n)$$

is bounded, and $s - k = \frac{n}{2} + \gamma$. For $0 < \gamma < 1$ the Sobolev–Morrey embedding yields

$$H^{\frac{n}{2}+\gamma}(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n).$$

Hence $D^\alpha u \in C^{0,\gamma}$ for every $|\alpha| = k$, which means exactly that $u \in C^{k,\gamma}(\mathbb{R}^n)$. Moreover

$$\|u\|_{C^{k,\gamma}} \lesssim \sum_{|\alpha|=k} \|D^\alpha u\|_{C^{0,\gamma}} \lesssim \sum_{|\alpha|=k} \|D^\alpha u\|_{H^{\frac{n}{2}+\gamma}} \lesssim \|u\|_{H^{\frac{n}{2}+k+\gamma}},$$

so the embedding is continuous. $\square$

Exercise 7

Fix $s \in \mathbb{R}$, and suppose that $f \in H^s(\mathbb{R}^n)$.

(a) Show that there exists a unique $u \in H^{s+4}(\mathbb{R}^n)$ solving

$$\Delta^2 u + u = f \quad \text{in } \mathbb{R}^n.$$

(b) Show that there exists a constant $C > 0$ such that

$$\|u\|_{H^{s+4}} \leq C \|f\|_{H^s}.$$

(c) For which values of s does the equation hold in the sense of classical derivatives (possibly after redefining u and f on sets of measure zero)?

Solution. Taking Fourier transforms gives

$$(1 + |\xi|^4) \widehat{u}(\xi) = \widehat{f}(\xi), \quad \widehat{u}(\xi) = m(\xi) \widehat{f}(\xi), \quad m(\xi) = \frac{1}{1 + |\xi|^4}.$$

The multiplier m satisfies

$$(1 + |\xi|^2)^{\frac{s+4}{2}} |m(\xi)| \lesssim (1 + |\xi|^2)^{\frac{s}{2}},$$

so

$$\|u\|_{H^{s+4}}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{s+4} |\widehat{u}(\xi)|^2 d\xi = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{s+4} |m(\xi)|^2 |\widehat{f}(\xi)|^2 d\xi \lesssim \|f\|_{H^s}^2.$$

Thus $u \in H^{s+4}(\mathbb{R}^n)$ and the bound in (b) holds. Uniqueness follows from the fact that $1 + |\xi|^4 \neq 0$: if $f = 0$, then $(1 + |\xi|^4)\widehat{u} = 0$ implies $\widehat{u} \equiv 0$, hence $u \equiv 0$.

For (c), recall that if $t > \frac{n}{2} + k + \alpha$ with $k \in \mathbb{N}$ and $0 < \alpha < 1$, then $H^t(\mathbb{R}^n) \hookrightarrow C^{k,\alpha}(\mathbb{R}^n)$. We have $u \in H^{s+4}(\mathbb{R}^n)$, and to obtain $u \in C^{4,\alpha}$ for some $\alpha > 0$ we require

$$s + 4 > \frac{n}{2} + 4 + \alpha \quad \text{for some } \alpha \in (0, 1),$$

which is equivalent to

$$s > \frac{n}{2}.$$

If $s > \frac{n}{2}$, then $f \in H^s(\mathbb{R}^n) \hookrightarrow C^{0,\alpha}$ and $u \in H^{s+4} \hookrightarrow C^{4,\alpha}$. After modifying u and f on sets of measure zero, the identity $\Delta^2 u + u = f$ holds pointwise with classical derivatives. For $s \leq \frac{n}{2}$ one generally only has the equation in the weak (distributional) sense. $\square$

Classification of Sobolev-type spaces

1. Classical integer-order spaces. For $k \in \mathbb{N}$, $1 \leq p \leq \infty$ and an open $\Omega \subset \mathbb{R}^n$,

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \leq k\},$$

with norm $\|u\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p}$. For $p = 2$ we write $H^k(\Omega) = W^{k,2}(\Omega)$.

2. Homogeneous vs. inhomogeneous. The homogeneous Sobolev spaces on $\mathbb{R}^n$ are

$$\dot{W}^{k,p}(\mathbb{R}^n) := \{u \in \mathcal{S}' : D^\alpha u \in L^p, |\alpha| = k\},$$

and in the Hilbert case

$$\dot{H}^s(\mathbb{R}^n) := \left\{ u \in \mathcal{S}' : \int_{\mathbb{R}^n} |\xi|^{2s} |\widehat{u}(\xi)|^2 d\xi < \infty \right\}.$$

They are scale-invariant counterparts of $W^{k,p}$ and H^s .

3. Fractional Sobolev spaces.

(a) *Bessel-potential spaces* (Hilbert case):

$$H^s(\mathbb{R}^n) := \left\{ u \in \mathcal{S}' : \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{u}(\xi)|^2 d\xi < \infty \right\},$$

equivalently $(I - \Delta)^{s/2} u \in L^2$.

(b) *Slobodeckij spaces* for $0 < s < 1$, $1 \leq p < \infty$:

$$W^{s,p}(\mathbb{R}^n) := \left\{ u \in L^p : [u]_{W^{s,p}}^p = \iint_{\mathbb{R}^n \times \mathbb{R}^n} \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} dx dy < \infty \right\}.$$

For $p = 2$ and suitable domains one has $W^{s,2} \simeq H^s$.

(c) *Negative-order spaces* are defined as duals: $W^{-s,p}(\Omega) := (W_0^{s,p'}(\Omega))'$ and $H^{-s}(\mathbb{R}^n) := (H^s(\mathbb{R}^n))'$.

4. Periodic Sobolev spaces. On the torus $\mathbb{T}^n$,

$$H^s(\mathbb{T}^n) = \left\{ u : \sum_{k \in \mathbb{Z}^n} (1 + |k|^2)^s |\hat{u}_k|^2 < \infty \right\},$$

and $W^{k,p}(\mathbb{T}^n)$ is defined via periodic derivatives.

5. Variants with additional structure.

- Weighted Sobolev spaces $W_{\omega}^{k,p}(\mathbb{R}^n)$ with Muckenhoupt weights ω .
- Anisotropic Sobolev spaces with different regularity in different directions.
- Sobolev spaces on manifolds, graphs, etc., defined via charts or spectral calculus.
- Subspaces with constraints, e.g. $H_0^1(\Omega)$, divergence-free spaces, etc.

6. Embedding into the Besov/Triebel–Lizorkin scale. Sobolev spaces are particular cases of Besov and Triebel–Lizorkin spaces:

$$H^s(\mathbb{R}^n) = F_{2,2}^s(\mathbb{R}^n), \quad W^{k,p}(\mathbb{R}^n) = F_{p,2}^k(\mathbb{R}^n),$$

with the more general scales $B_{p,q}^s$ and $F_{p,q}^s$ obtained from Littlewood–Paley decompositions.

Exercise 8 (Trace theorem on a hyperplane)

Assume $s > \frac{1}{2}$ and suppose $u \in \mathcal{S}(\mathbb{R}^n)$. Write $x = (x', x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and define the trace

$$Tu(x') := u(x', 0), \quad x' \in \mathbb{R}^{n-1}.$$

(a)

Show that for each $\xi' \in \mathbb{R}^{n-1}$ one has

$$\widehat{Tu}(\xi') = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{u}(\xi', \xi_n) d\xi_n.$$

(b)

Deduce that

$$\left| \widehat{Tu}(\xi') \right|^2 \leq \frac{1}{(2\pi)^2} \left(\int_{\mathbb{R}} (1 + |\xi|^2)^s |\widehat{u}(\xi', \xi_n)|^2 d\xi_n \right) \left(\int_{\mathbb{R}} \frac{d\xi_n}{(1 + |\xi|^2)^s} \right),$$

where $\xi = (\xi', \xi_n)$.

(c)

By changing variables in the second integral above to $\xi_n = t\sqrt{1 + |\xi'|^2}$, show that there exists a constant $C(s) > 0$ such that

$$\|Tu\|_{H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})} \leq C(s) \|u\|_{H^s(\mathbb{R}^n)}.$$

(d)

Conclude that T extends uniquely by density to a bounded linear operator

$$T : H^s(\mathbb{R}^n) \longrightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1}), \quad s > \frac{1}{2}.$$

(e)

Suppose $\nu \in \mathcal{S}(\mathbb{R}^{n-1})$ and let $\phi \in C_c^\infty(\mathbb{R})$ satisfy

$$\int_{\mathbb{R}} \phi(t) dt = \sqrt{2\pi}.$$

Define u through its Fourier transform by

$$\widehat{u}(\xi', \xi_n) := \frac{\widehat{\nu}(\xi')}{\sqrt{1 + |\xi'|^2}} \phi\left(\frac{\xi_n}{\sqrt{1 + |\xi'|^2}}\right).$$

Show that there exists a constant $C > 0$ such that

$$\|u\|_{H^s(\mathbb{R}^n)} \leq C \|\nu\|_{H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})}.$$

Compute Tu and verify that $Tu = \nu$. Conclude that

$$T : H^s(\mathbb{R}^n) \longrightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})$$

is surjective and hence an isomorphism onto $H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})$.

Theoretical background. For $s > \frac{1}{2}$ the trace operator

$$T : H^s(\mathbb{R}^n) \rightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1}), \quad Tu(x') = u(x', 0),$$

is well defined, bounded and surjective. The proof above is based on the Fourier identity in (a), the Cauchy–Schwarz estimate in (b), and the estimate

$$\int_{\mathbb{R}} \frac{d\xi_n}{(1 + |\xi'|^2 + \xi_n^2)^s} = c_s (1 + |\xi'|^2)^{\frac{1}{2}-s}, \quad s > \frac{1}{2},$$

which exactly produces the $H^{s-\frac{1}{2}}$ -weight in ξ' . Part (e) constructs a bounded extension operator $E : H^{s-\frac{1}{2}}(\mathbb{R}^{n-1}) \rightarrow H^s(\mathbb{R}^n)$ with $T \circ E = \text{Id}$, showing that every element of $H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})$ arises as a trace.

Exercise 8 (solutions)

Let $s > \frac{1}{2}$ and $u \in \mathcal{S}(\mathbb{R}^n)$, $x = (x', x_n)$, $\xi = (\xi', \xi_n)$.

(a)

We have

$$\widehat{Tu}(\xi') = \int_{\mathbb{R}^{n-1}} e^{-ix' \cdot \xi'} u(x', 0) dx'.$$

Using the inverse Fourier representation of u ,

$$u(x', 0) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} e^{i(x' \cdot \eta' + 0 \cdot \eta_n)} \widehat{u}(\eta', \eta_n) d\eta_n d\eta',$$

we obtain

$$\widehat{Tu}(\xi') = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} \left(\int_{\mathbb{R}^{n-1}} e^{-ix' \cdot \xi'} e^{ix' \cdot \eta'} dx' \right) \widehat{u}(\eta', \eta_n) d\eta_n d\eta'.$$

The inner integral equals $(2\pi)^{n-1} \delta(\eta' - \xi')$, hence

$$\widehat{Tu}(\xi') = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{u}(\xi', \xi_n) d\xi_n.$$

(b)

From (a),

$$\widehat{Tu}(\xi') = \frac{1}{2\pi} \int_{\mathbb{R}} (1 + |\xi|^2)^{\frac{s}{2}} \widehat{u}(\xi', \xi_n) (1 + |\xi|^2)^{-\frac{s}{2}} d\xi_n.$$

By the Cauchy–Schwarz inequality,

$$|\widehat{Tu}(\xi')|^2 \leq \frac{1}{(2\pi)^2} \left(\int_{\mathbb{R}} (1 + |\xi|^2)^s |\widehat{u}(\xi', \xi_n)|^2 d\xi_n \right) \left(\int_{\mathbb{R}} \frac{d\xi_n}{(1 + |\xi|^2)^s} \right).$$

(c)

Set

$$I(\xi') := \int_{\mathbb{R}} \frac{d\xi_n}{(1 + |\xi'|^2 + \xi_n^2)^s}.$$

With $L = \sqrt{1 + |\xi'|^2}$ and $\xi_n = Lt$,

$$I(\xi') = L^{1-2s} \int_{\mathbb{R}} \frac{dt}{(1 + t^2)^s} = C_s (1 + |\xi'|^2)^{\frac{1}{2}-s},$$

where $C_s < \infty$ for $s > \frac{1}{2}$. Using (b),

$$\begin{aligned} \|Tu\|_{H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})}^2 &= \int_{\mathbb{R}^{n-1}} (1 + |\xi'|^2)^{s-\frac{1}{2}} |\widehat{Tu}(\xi')|^2 d\xi' \\ &\leq \frac{1}{(2\pi)^2} \int_{\mathbb{R}^{n-1}} (1 + |\xi'|^2)^{s-\frac{1}{2}} \left(\int_{\mathbb{R}} (1 + |\xi|^2)^s |\widehat{u}(\xi', \xi_n)|^2 d\xi_n \right) I(\xi') d\xi' \\ &= \frac{C_s}{(2\pi)^2} \iint_{\mathbb{R}^{n-1} \times \mathbb{R}} (1 + |\xi|^2)^s |\widehat{u}(\xi', \xi_n)|^2 d\xi_n d\xi' \\ &= C(s) \|u\|_{H^s(\mathbb{R}^n)}^2. \end{aligned}$$

Thus $\|Tu\|_{H^{s-\frac{1}{2}}} \leq C(s) \|u\|_{H^s}$.

(d)

The map $T : \mathcal{S}(\mathbb{R}^n) \rightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})$ is bounded by (c). Since $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$, T extends uniquely by continuity to a bounded linear operator

$$T : H^s(\mathbb{R}^n) \longrightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1}), \quad s > \frac{1}{2}.$$

(e)

Let $\nu \in \mathcal{S}(\mathbb{R}^{n-1})$ and $\phi \in C_c^\infty(\mathbb{R})$ satisfy $\int_{\mathbb{R}} \phi(t) dt = \sqrt{2\pi}$. Define

$$\widehat{u}(\xi', \xi_n) := \frac{\widehat{\nu}(\xi')}{\sqrt{1 + |\xi'|^2}} \phi\left(\frac{\xi_n}{\sqrt{1 + |\xi'|^2}}\right).$$

Then

$$\begin{aligned} \|u\|_{H^s(\mathbb{R}^n)}^2 &= \iint_{\mathbb{R}^{n-1} \times \mathbb{R}} (1 + |\xi|^2)^s |\widehat{u}(\xi', \xi_n)|^2 d\xi_n d\xi' \\ &= \int_{\mathbb{R}^{n-1}} \frac{|\widehat{\nu}(\xi')|^2}{1 + |\xi'|^2} \left(\int_{\mathbb{R}} (1 + |\xi'|^2 + \xi_n^2)^s \left| \phi\left(\frac{\xi_n}{\sqrt{1 + |\xi'|^2}}\right) \right|^2 d\xi_n \right) d\xi'. \end{aligned}$$

With $L = \sqrt{1 + |\xi'|^2}$ and $\xi_n = Lt$,

$$\int_{\mathbb{R}} (1 + |\xi'|^2 + \xi_n^2)^s \left| \phi\left(\frac{\xi_n}{L}\right) \right|^2 d\xi_n = L^{2s+1} \int_{\mathbb{R}} (1 + t^2)^s |\phi(t)|^2 dt \leq C_\phi (1 + |\xi'|^2)^{s+\frac{1}{2}},$$

where $C_\phi := \int_{\mathbb{R}} (1+t^2)^s |\phi(t)|^2 dt$. Hence

$$\|u\|_{H^s(\mathbb{R}^n)}^2 \leq C_\phi \int_{\mathbb{R}^{n-1}} |\hat{\nu}(\xi')|^2 (1+|\xi'|^2)^{s-\frac{1}{2}} d\xi' = C_\phi \|\nu\|_{H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})}^2.$$

Using (a),

$$\widehat{Tu}(\xi') = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{u}(\xi', \xi_n) d\xi_n = \frac{\hat{\nu}(\xi')}{2\pi\sqrt{1+|\xi'|^2}} \int_{\mathbb{R}} \phi\left(\frac{\xi_n}{\sqrt{1+|\xi'|^2}}\right) d\xi_n.$$

With $\xi_n = Lt$, $L = \sqrt{1+|\xi'|^2}$,

$$\int_{\mathbb{R}} \phi\left(\frac{\xi_n}{L}\right) d\xi_n = L \int_{\mathbb{R}} \phi(t) dt = L\sqrt{2\pi}.$$

Thus

$$\widehat{Tu}(\xi') = \frac{\hat{\nu}(\xi')}{2\pi\sqrt{1+|\xi'|^2}} L\sqrt{2\pi} = \hat{\nu}(\xi'),$$

so $Tu = \nu$. By density, this defines a bounded extension operator $E : H^{s-\frac{1}{2}}(\mathbb{R}^{n-1}) \rightarrow H^s(\mathbb{R}^n)$ with $T \circ E = \text{Id}$. Therefore $T : H^s(\mathbb{R}^n) \rightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})$ is surjective. $\square$

Exercise 9 — Dirichlet energy and eigenvalues

Let $\Omega \subset \mathbb{R}^n$ be open and bounded. For $u \in H_0^1(\Omega)$ define the Dirichlet energy

$$E[u] = \int_{\Omega} |Du(x)|^2 dx.$$

Problem statement

- (a) Suppose that $(u_i)_{i=1}^\infty$ is a sequence with $u_i \in H_0^1(\Omega)$ and $u_i \rightharpoonup u$ weakly in $H_0^1(\Omega)$. Show that

$$E[u] \leq \liminf_{i \rightarrow \infty} E[u_i].$$

- (b) Consider the set

$$\mathcal{E}_1 = \{E[u] : u \in H_0^1(\Omega), \|u\|_{L^2} = 1\}.$$

Let $\lambda_1 := \inf \mathcal{E}_1$. Show that there exists $w_1 \in H_0^1(\Omega)$ with $\|w_1\|_{L^2} = 1$ and $E[w_1] = \lambda_1$, and deduce that $\lambda_1 > 0$.

- (c) Deduce that

$$\lambda_1 \|u\|_{L^2}^2 \leq \int_{\Omega} |Du|^2 dx$$

holds for all $u \in H_0^1(\Omega)$, with equality for $u = w_1$. This is Poincaré's inequality.

- (d) By considering $u = w_1 + t\varphi$ for $t \in \mathbb{R}$ and $\varphi \in \mathcal{D}(\Omega) = C_c^\infty(\Omega)$, or otherwise, show that w_1 satisfies

$$-\Delta w_1 = \lambda_1 w_1$$

in the sense of distributions, i.e. in $\mathcal{D}'(\Omega)$.

- (e) Suppose $\chi \in C_c^\infty(\Omega)$ and put $v = \chi w_1$. Show that v satisfies

$$-\Delta v + v = f$$

in $\mathcal{S}'(\mathbb{R}^n)$, where $f \in L^2(\mathbb{R}^n)$. Deduce that $v \in H^2(\mathbb{R}^n)$. By iterating this argument, deduce that

$$w_1 \in H_0^1(\Omega) \cap C^\infty(\Omega).$$

- (f) Consider

$$\mathcal{E}_2 = \{E[u] : u \in H_0^1(\Omega), \|u\|_{L^2} = 1, (u, w_1)_{L^2} = 0\}.$$

Show that there exist $\lambda_2 \geq \lambda_1$ and $w_2 \in H_0^1(\Omega) \cap C^\infty(\Omega)$ with $w_2 \neq w_1$, $\|w_2\|_{L^2} = 1$ and

$$-\Delta w_2 = \lambda_2 w_2.$$

Theory

The space $H_0^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in the Sobolev norm

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|Du\|_{L^2}^2.$$

The functional $E[u] = \|Du\|_{L^2}^2$ is convex and weakly lower semicontinuous on $H_0^1(\Omega)$. Since Ω is bounded, the embedding $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact (Rellich–Kondrachov).

The Dirichlet Laplacian $-\Delta$ with domain $H_0^1(\Omega)$ has discrete spectrum

$$0 < \lambda_1 \leq \lambda_2 \leq \cdots \rightarrow \infty,$$

and the first eigenvalue admits the variational characterisation

$$\lambda_1 = \inf \left\{ \frac{\int_\Omega |Du|^2}{\int_\Omega |u|^2} : u \in H_0^1(\Omega), u \neq 0 \right\}.$$

Minimisers are eigenfunctions solving $-\Delta u = \lambda_1 u$.

For the constant-coefficient elliptic operator $-\Delta + I$ on $\mathbb{R}^n$ the equation $(-\Delta + I)v = f$ with $f \in L^2(\mathbb{R}^n)$ implies $v \in H^2(\mathbb{R}^n)$. By iterating and using Sobolev embedding, one derives interior smoothness for solutions of $-\Delta u = \lambda u$.

Solution

- (a) Since $u_i \rightharpoonup u$ in $H_0^1(\Omega)$, one has $Du_i \rightharpoonup Du$ in $L^2(\Omega; \mathbb{R}^n)$. The map $L^2 \ni f \mapsto \|f\|_{L^2}^2$ is convex and weakly lower semicontinuous, hence

$$E[u] = \|Du\|_{L^2}^2 \leq \liminf_{i \rightarrow \infty} \|Du_i\|_{L^2}^2 = \liminf_{i \rightarrow \infty} E[u_i].$$

(b) By definition of λ_1 there exists a sequence $(u_k) \subset H_0^1(\Omega)$ with

$$\|u_k\|_{L^2} = 1, \quad E[u_k] \rightarrow \lambda_1.$$

Then

$$\|u_k\|_{H^1}^2 = \|u_k\|_{L^2}^2 + \|Du_k\|_{L^2}^2 = 1 + E[u_k],$$

so (u_k) is bounded in $H_0^1(\Omega)$. By weak compactness and the Rellich–Kondrachov theorem there is a subsequence (denoted again u_k) and $w_1 \in H_0^1(\Omega)$ such that

$$u_k \rightharpoonup w_1 \quad \text{in } H_0^1(\Omega), \quad u_k \rightarrow w_1 \quad \text{in } L^2(\Omega).$$

In particular

$$\|w_1\|_{L^2} = \lim_{k \rightarrow \infty} \|u_k\|_{L^2} = 1.$$

By part (a),

$$E[w_1] \leq \liminf_{k \rightarrow \infty} E[u_k] = \lambda_1.$$

Since λ_1 is the infimum of $\mathcal{E}_1$, we also have $\lambda_1 \leq E[w_1]$. Thus $E[w_1] = \lambda_1$.

If $\lambda_1 = 0$, then $E[w_1] = 0$ and hence $Dw_1 = 0$ a.e., so w_1 is constant. Because $w_1 \in H_0^1(\Omega)$ its trace on $\partial\Omega$ is zero, hence $w_1 \equiv 0$, contradicting $\|w_1\|_{L^2} = 1$. Therefore $\lambda_1 > 0$.

(c) Let $u \in H_0^1(\Omega)$, $u \neq 0$, and set $v = u/\|u\|_{L^2}$. Then $\|v\|_{L^2} = 1$ and by definition of λ_1

$$\lambda_1 \leq E[v] = \int_{\Omega} |Dv|^2 = \frac{1}{\|u\|_{L^2}^2} \int_{\Omega} |Du|^2.$$

Hence

$$\lambda_1 \|u\|_{L^2}^2 \leq \int_{\Omega} |Du|^2.$$

For $u = 0$ the inequality holds trivially. If $u = w_1$, then $E[u] = \lambda_1 \|u\|_{L^2}^2$ and equality holds.

(d) Consider the functional

$$J(u) := E[u] - \lambda_1 \|u\|_{L^2}^2.$$

Since w_1 minimises E under the constraint $\|u\|_{L^2} = 1$, there is a Lagrange multiplier λ_1 and w_1 is a stationary point of J . For $\varphi \in C_c^\infty(\Omega)$ one has

$$0 = \left. \frac{d}{dt} J(w_1 + t\varphi) \right|_{t=0} = 2 \int_{\Omega} Dw_1 \cdot D\varphi - 2\lambda_1 \int_{\Omega} w_1 \varphi.$$

Thus

$$\int_{\Omega} Dw_1 \cdot D\varphi = \lambda_1 \int_{\Omega} w_1 \varphi, \quad \forall \varphi \in C_c^\infty(\Omega).$$

By the definition of the distributional Laplacian,

$$\langle -\Delta w_1, \varphi \rangle = \int_{\Omega} Dw_1 \cdot D\varphi,$$

so

$$\langle -\Delta w_1, \varphi \rangle = \lambda_1 \int_{\Omega} w_1 \varphi = \langle \lambda_1 w_1, \varphi \rangle$$

for all $\varphi \in C_c^\infty(\Omega)$, which means

$$-\Delta w_1 = \lambda_1 w_1 \quad \text{in } \mathcal{D}'(\Omega).$$

(e) Let $\chi \in C_c^\infty(\Omega)$ and put $v = \chi w_1$, extended by zero to $\mathbb{R}^n$. Then

$$-\Delta v = -\Delta(\chi w_1) = -\chi \Delta w_1 - 2\nabla \chi \cdot \nabla w_1 - w_1 \Delta \chi.$$

Using $-\Delta w_1 = \lambda_1 w_1$ we obtain

$$-\Delta v + v = \chi(\lambda_1 + 1)w_1 - 2\nabla \chi \cdot \nabla w_1 - w_1 \Delta \chi =: f.$$

Since $w_1 \in H_0^1(\Omega)$ and $\chi, \nabla \chi, \Delta \chi$ are smooth with compact support, we have $f \in L^2(\mathbb{R}^n)$.

Taking Fourier transforms on $\mathbb{R}^n$, the equation $-\Delta v + v = f$ becomes

$$(|\xi|^2 + 1)\hat{v}(\xi) = \hat{f}(\xi).$$

Thus

$$\hat{v}(\xi) = \frac{\hat{f}(\xi)}{|\xi|^2 + 1}$$

and

$$\int_{\mathbb{R}^n} (1 + |\xi|^2)^2 |\hat{v}(\xi)|^2 d\xi \leq \int_{\mathbb{R}^n} |\hat{f}(\xi)|^2 d\xi < \infty,$$

so $v \in H^2(\mathbb{R}^n)$. Since χ was arbitrary, this shows that $w_1 \in H_{\text{loc}}^2(\Omega)$. Iterating the argument gives $w_1 \in H_{\text{loc}}^k(\Omega)$ for all k , and therefore $w_1 \in C^\infty(\Omega)$ by Sobolev embedding. Together with $w_1 \in H_0^1(\Omega)$ we obtain

$$w_1 \in H_0^1(\Omega) \cap C^\infty(\Omega).$$

(f) Define

$$\mathcal{E}_2 = \{E[u] : u \in H_0^1(\Omega), \|u\|_{L^2} = 1, (u, w_1)_{L^2} = 0\}, \quad \lambda_2 := \inf \mathcal{E}_2.$$

Choose a sequence (u_k) with $\|u_k\|_{L^2} = 1$, $(u_k, w_1)_{L^2} = 0$ and $E[u_k] \rightarrow \lambda_2$. As in (b), (u_k) is bounded in $H_0^1(\Omega)$, so up to a subsequence

$$u_k \rightharpoonup w_2 \text{ in } H_0^1(\Omega), \quad u_k \rightarrow w_2 \text{ in } L^2(\Omega).$$

Then

$$\|w_2\|_{L^2} = 1, \quad (w_2, w_1)_{L^2} = 0,$$

and by part (a),

$$E[w_2] \leq \liminf_{k \rightarrow \infty} E[u_k] = \lambda_2.$$

Since λ_2 is the infimum over the same constraint set, we must have $E[w_2] = \lambda_2$.

For every $u \in H_0^1(\Omega)$ with $u \neq 0$ we have

$$\frac{E[u]}{\|u\|_{L^2}^2} \geq \lambda_1,$$

so in particular $\lambda_2 \geq \lambda_1$.

To derive the Euler–Lagrange equation, consider

$$J(u) = E[u] - \lambda_2 \|u\|_{L^2}^2 - \mu(u, w_1)_{L^2},$$

with Lagrange multipliers λ_2, μ . Stationarity of J at w_2 in the direction $\varphi \in C_c^\infty(\Omega)$ gives

$$\int_{\Omega} Dw_2 \cdot D\varphi = \lambda_2 \int_{\Omega} w_2 \varphi + \frac{\mu}{2} \int_{\Omega} w_1 \varphi.$$

Choosing $\varphi = w_1$ shows $\mu = 0$ (using $(w_2, w_1)_{L^2} = 0$), and thus

$$\int_{\Omega} Dw_2 \cdot D\varphi = \lambda_2 \int_{\Omega} w_2 \varphi, \quad \forall \varphi \in C_c^\infty(\Omega),$$

which means

$$-\Delta w_2 = \lambda_2 w_2 \quad \text{in } \mathcal{D}'(\Omega).$$

The same localisation and regularity argument as in (e) shows that $w_2 \in C^\infty(\Omega)$, so $w_2 \in H_0^1(\Omega) \cap C^\infty(\Omega)$.

Example: the interval $(0, \pi)$

Take $\Omega = (0, \pi) \subset \mathbb{R}$. Then $H_0^1(0, \pi)$ consists of L^2 -functions with square-integrable weak derivative and zero trace at 0 and π .

The eigenfunctions of the Dirichlet Laplacian are

$$w_k(x) = \sqrt{\frac{2}{\pi}} \sin(kx), \quad k = 1, 2, \dots,$$

and they satisfy

$$-w_k'' = k^2 w_k, \quad \|w_k\|_{L^2(0, \pi)} = 1.$$

Hence the eigenvalues are $\lambda_k = k^2$. In particular, $\lambda_1 = 1$ and Poincaré's inequality becomes

$$\int_0^\pi |u(x)|^2 dx \leq \int_0^\pi |u'(x)|^2 dx, \quad u \in H_0^1(0, \pi),$$

with equality for multiples of $\sin x$.

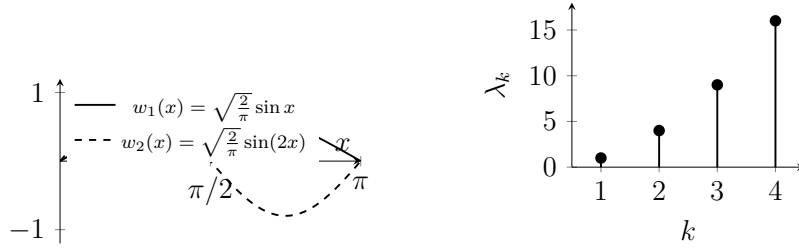


Figure 14: Left: first two Dirichlet eigenfunctions on $(0, \pi)$. Right: the first few eigenvalues $\lambda_k = k^2$.

Plots for the one-dimensional example

On the interval $\Omega = (0, \pi)$ the normalised Dirichlet eigenfunctions and eigenvalues are

$$w_k(x) = \sqrt{\frac{2}{\pi}} \sin(kx), \quad -w_k'' = \lambda_k w_k, \quad \lambda_k = k^2, \quad k = 1, 2, \dots$$

The first eigenfunction w_1 has a single bump and no interior zero, while w_2 has two bumps and one interior zero at $x = \pi/2$. Both vanish at the endpoints, corresponding to the Dirichlet boundary conditions. The associated eigenvalues form a sequence of “energy levels” $1, 4, 9, 16, \dots$ which grow quadratically with k .

To visualise this, it is convenient to place a plot of the first two eigenfunctions next to a simple diagram of the first few eigenvalues. Using `pgfplots` one may use the following code.¹

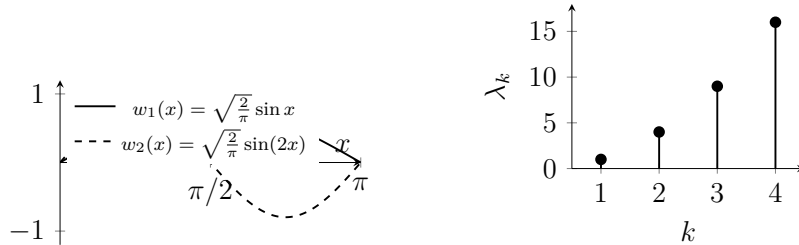


Figure 15: Left: first two Dirichlet eigenfunctions on $(0, \pi)$. Right: the first few eigenvalues $\lambda_k = k^2$.

Spectrum and eigenvalue sequence

On the interval $(0, \pi)$ the Dirichlet Laplacian is the operator

$$A = -\frac{d^2}{dx^2}, \quad D(A) = \{u \in H^2(0, \pi) \cap H_0^1(0, \pi)\}.$$

¹Make sure to load `\usepackage{pgfplots}` and set a compatible version, e.g. `\pgfplotsset{compat=1.18}`.

Viewed as an operator on $L^2(0, \pi)$ it is self-adjoint and positive. Its spectrum consists entirely of eigenvalues and is given by

$$\sigma(A) = \{\lambda_k : k \in \mathbb{N}\}, \quad \lambda_k = k^2.$$

There is no continuous spectrum. The associated eigenfunctions w_k form an orthonormal basis of $L^2(0, \pi)$, so every $u \in L^2(0, \pi)$ admits the expansion

$$u(x) = \sum_{k=1}^{\infty} c_k w_k(x), \quad c_k = (u, w_k)_{L^2(0, \pi)}.$$

For such u in the domain of A one has

$$Au = -u'' = \sum_{k=1}^{\infty} \lambda_k c_k w_k$$

and the Dirichlet energy may be written in spectral form as

$$E[u] = \int_0^\pi |u'(x)|^2 dx = \sum_{k=1}^{\infty} \lambda_k |c_k|^2.$$

In the general case of a bounded domain $\Omega \subset \mathbb{R}^n$ with sufficiently regular boundary, the Dirichlet Laplacian

$$A = -\Delta, \quad D(A) = \{u \in H_0^1(\Omega) : \Delta u \in L^2(\Omega)\},$$

is again self-adjoint and positive on $L^2(\Omega)$. Its spectrum consists of a discrete sequence of eigenvalues

$$0 < \lambda_1 \leq \lambda_2 \leq \cdots, \quad \lambda_k \rightarrow \infty \text{ as } k \rightarrow \infty,$$

each with finite multiplicity, and an orthonormal basis of $L^2(\Omega)$ made of smooth eigenfunctions. The first eigenvalue admits the Rayleigh–Ritz characterisation

$$\lambda_1 = \inf_{u \in H_0^1(\Omega) \setminus \{0\}} \frac{\int_{\Omega} |Du|^2 dx}{\int_{\Omega} |u|^2 dx},$$

and the strict inequality $\lambda_1 > 0$ expresses Poincaré’s inequality as a spectral gap between 0 and the bottom of the spectrum.